

Project Management

11. Quality Management & Close Project

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I CAN'T
REMEMBER IF I USED
CHOCOLATE CHIPS OR
MARBLES, BUT I GUESS
IT'S OK. SOMEONE'S SURE
TO FIGURE IT OUT BEFORE
THE BIG BAKE SALE.



Quality Management

It's not enough to make sure you get it done on time and under budget.

What is Quality Management

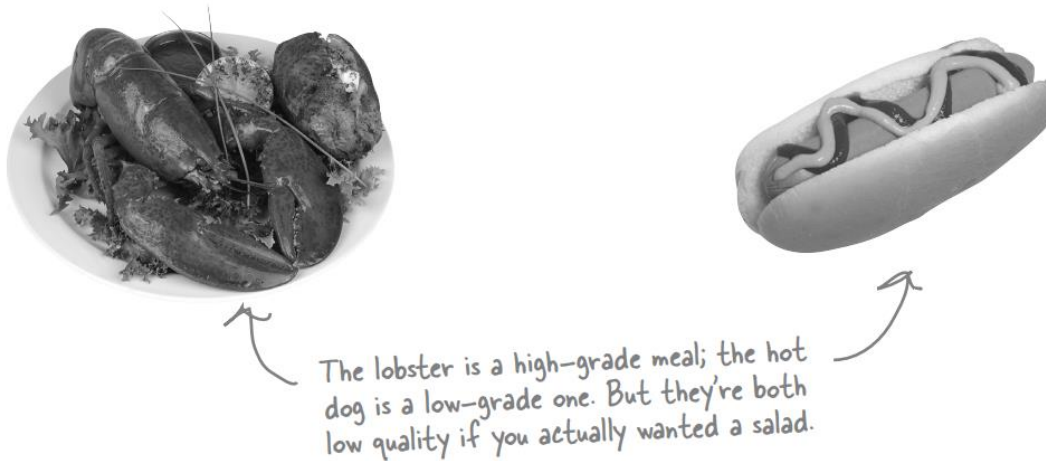
- Quality: the degree to which a product, service, or process **meets specified requirements** and **satisfies stakeholder expectations**.
 - Making sure that you build what you said you would, and that you do it as efficiently as you can.
- Project Quality Management includes the processes for incorporating the organization's quality policy regarding planning, managing, and controlling project and product quality requirements in order to meet stakeholders' objectives.



Quality vs. Grade

Quality means that something does **what you needed** it to do. **Grade** describes **how much** people value it.

- **Grade** is a category assigned to products or services having the same functional use but different technical characteristics.
- **Low Quality is always a problem; low grade may not be.**



The lobster is a high-grade meal; the hot dog is a low-grade one. But they're both low quality if you actually wanted a salad.

QA vs. QC

Quality Assurance

- QA deal with the **process**
- QA is for entire life cycle
- QA is **preventive** process
- QA make sure we are doing right things, the right way
- QA focuses on building in quality and hence preventing the defects

Quality Control

- QC deals with the **products**
- QC is for testing part
- QC is **corrective** process
- QC make sure the results of what we've done as expected
- QC focuses on testing of quality and hence detecting defects



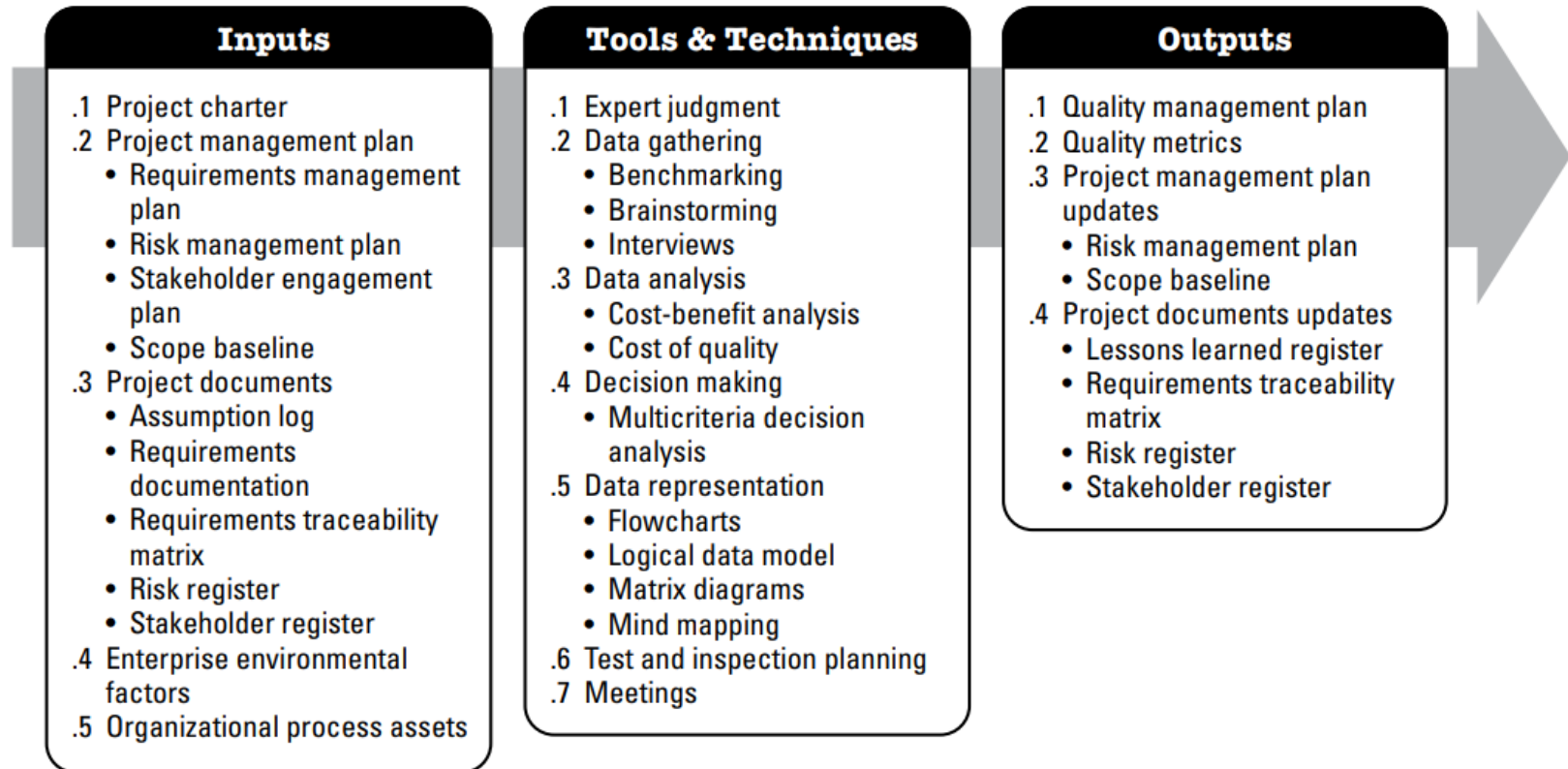
*When it comes to defects, **prevention is always better than inspection!***

Plan Quality Management

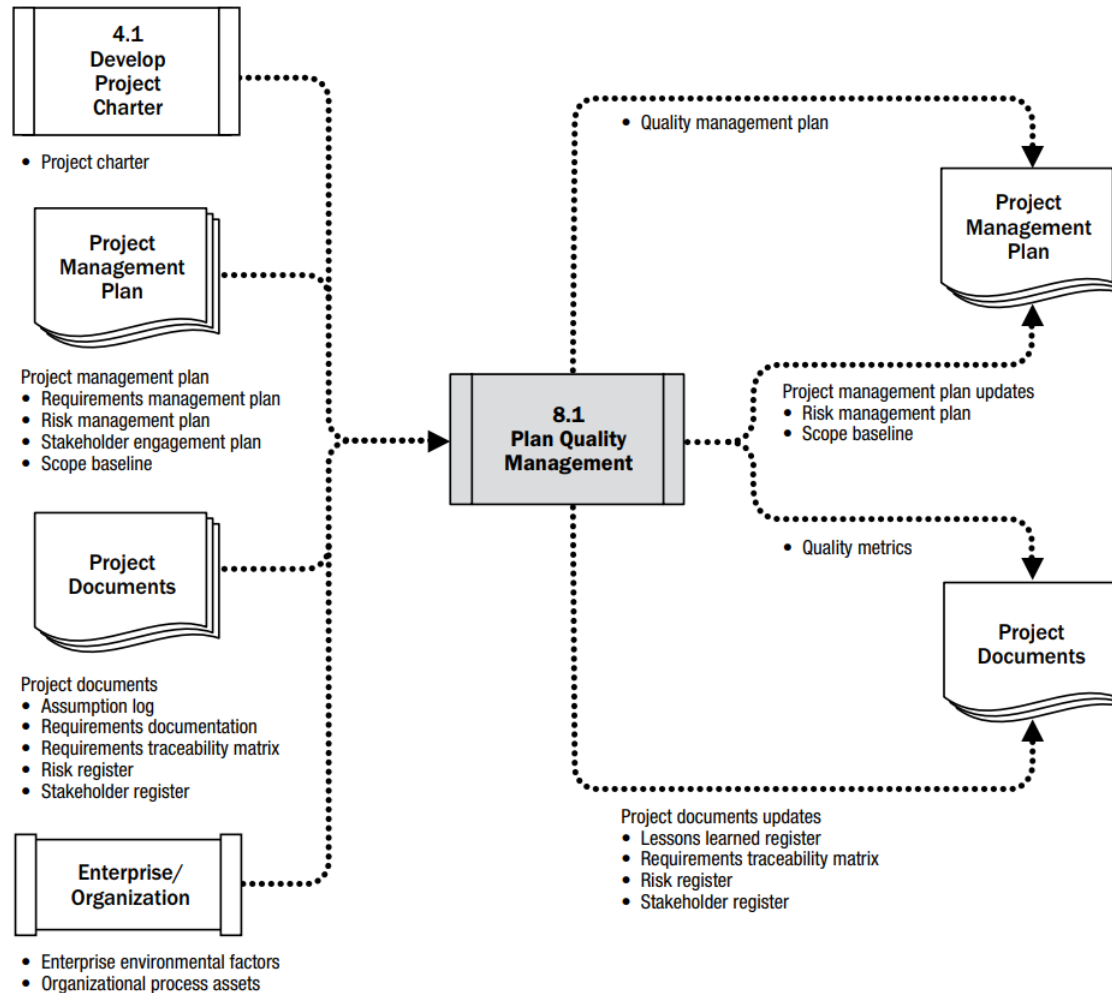
Plan Quality Management

- **What:** process of identifying quality requirements and/or standards for the project and its deliverables, and documenting how the project will demonstrate compliance with quality requirements and/or standards.
- **Why:** provides guidance and direction on how quality will be managed and verified throughout the project.
- **When:** performed once or at predefined points in the project.
 - Quality planning should be performed in parallel with the other planning processes.

Process



Data Flow



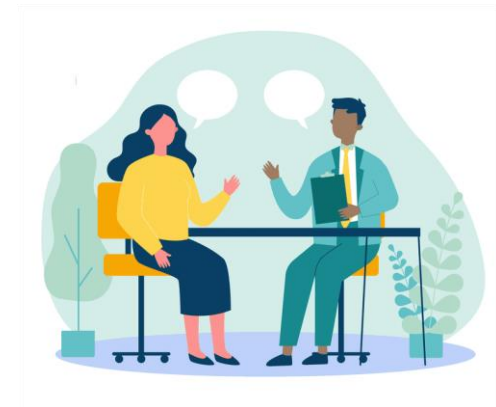
Tools & Techniques: Data-gathering

Benchmarking means using the results of Plan Quality on other projects to set goals for your own. Benchmarks can give you some reference points for judging your own project before you even get started with the work.

Brainstorming: Teams use brainstorming to identify the best way to manage quality on the project.

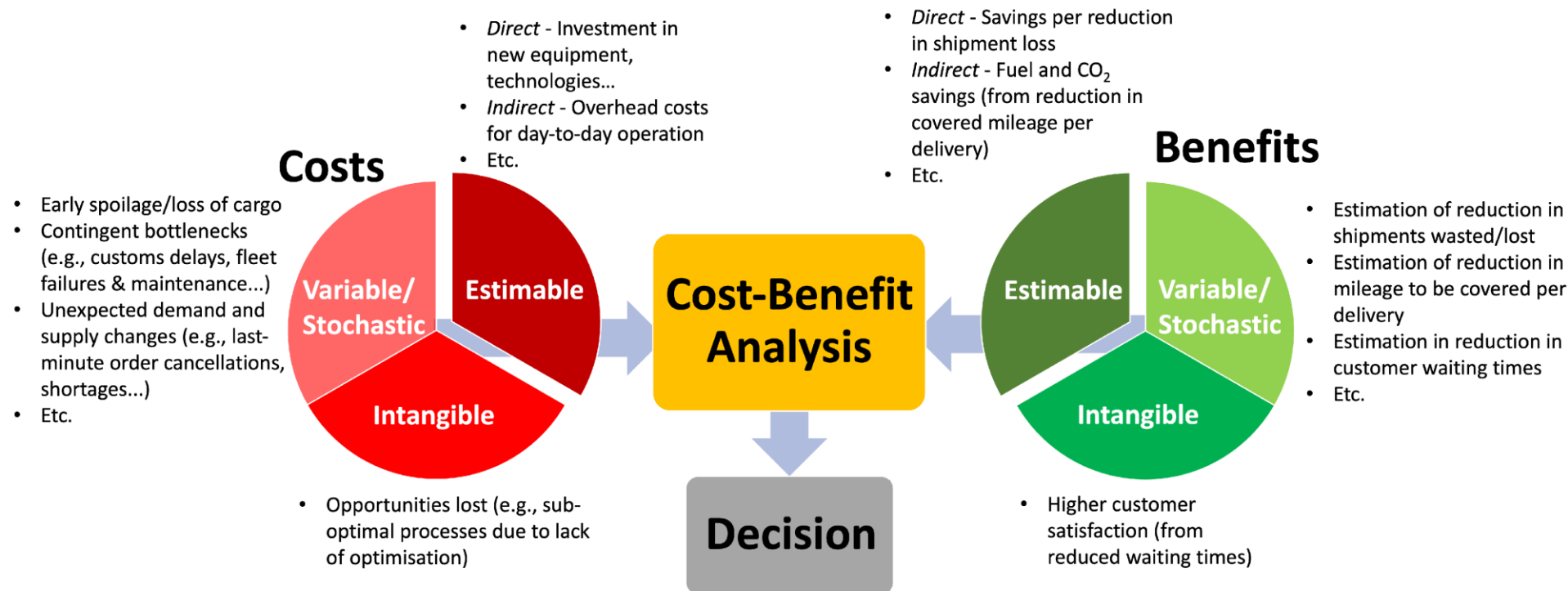
Interviews: The people participating in the project have a lot of information from their experience that can help you plan quality management.

				
	Hyundai Venue S Diesel	Mahindra XUV300 V4i Diesel	Maruti Vitara Brezza VXI	Tata Nexon XE Diesel
Price (On Road)	₹10,00,000	₹9,90,000	₹9,45,000	₹9,60,000
Fuel Type	Diesel	Diesel	Petrol	Diesel
Displacement(cc)	1493	1497	1462	1497
Max Power(bhp)	98.6	115	103.2	108.5
Mileage(kmpl)	23.7	20.0	17.03	21.5
Boot Space(lt.)	350	259	328	350
Transmission	Manual	Manual	Manual	Manual
Airbags	2	2	2	2
NCAP rating	4	5	4	5
Ground Clearance	190mm	180mm	200mm	209mm



Tools & Techniques: Data analysis

Cost-benefit analysis is looking at how much your quality activities will cost versus how much you will gain from doing them. Quality activities don't actually produce a product so it is harder for people to measure the benefits sometimes.

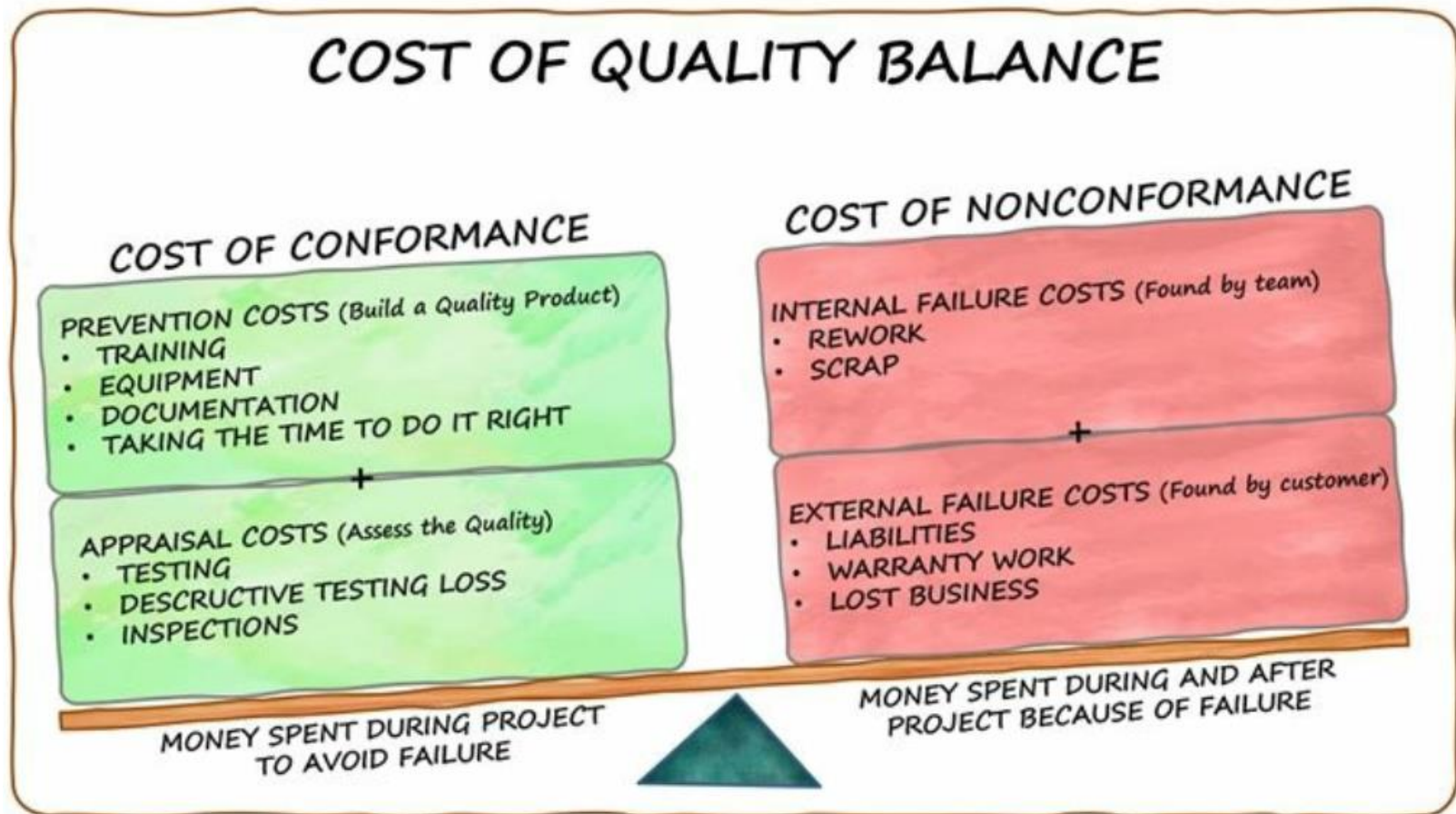


Net Benefit = Total Benefits – Total Costs

Benefit-Cost Ratio (BCR) = Total Benefits / Total Costs

Tools & Techniques: Data analysis

Cost of quality is what you get when you add up the *cost of all of the prevention and inspection activities* you are going to do on your project.



CoQ Analysis

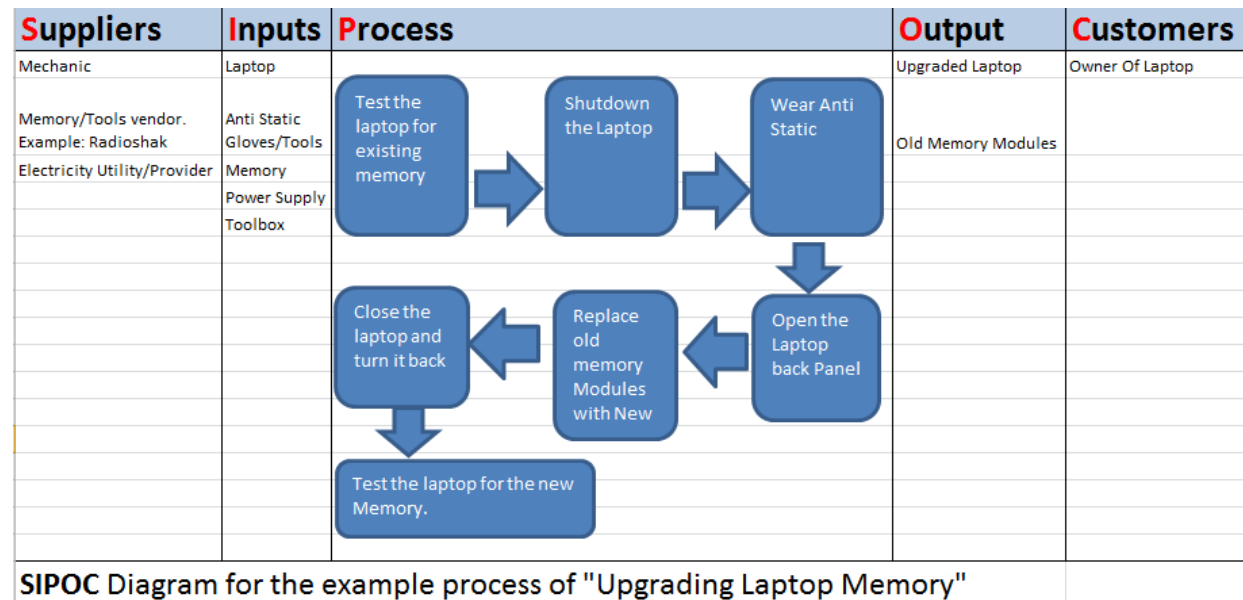
Type	What it represents	Timing	Goal
Prevention Costs	Cost to avoid defects (e.g. training, process design)	Before errors	Prevent errors from happening
Appraisal Costs	Cost to inspect/test quality	During process	Detect errors early
Internal Failure Costs	Cost to fix defects before delivery	Found before release	Minimize damage
External Failure Costs	Cost when defects reach the customer	Found after release	Recover reputation, fix product

Total CoQ = Prevention + Appraisal + Internal Failure + External Failure

Tools & Techniques: Data presentation

Flowcharts: showing the relationships among the process steps.

- Represent the steps in a process
- Help in understanding and estimating the cost of quality for a process.
- Identify where quality defects can occur or where to incorporate quality checks.
- As well as for process improvement

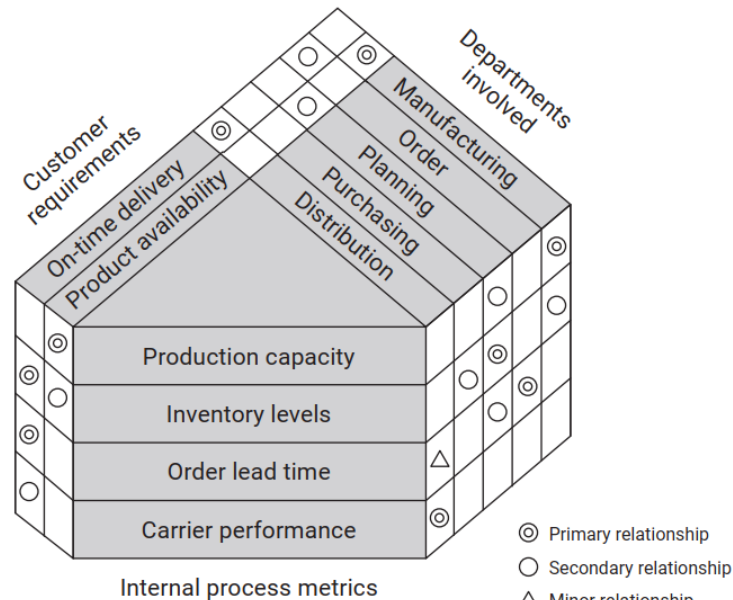


Tools & Techniques: Data presentation

Matrix diagram:

- Show the relationship between factors, causes and objectives in the rows and columns
- Use to **identify the key quality metrics** that are important for the success of the project.

Responsibilities for Performance to Customer Requirements



Tools & Techniques

Test and inspection planning

- Project manager and Project team determine **how to test or inspect** the product, deliverable, or service to meet the stakeholders' needs and expectations, as well as how to meet the goal for the product's performance and reliability.
- The tests and inspections are industry dependent. Ex: alpha and beta tests in software projects, strength tests in construction projects, inspection in manufacturing, and field tests and nondestructive tests in engineering.



Feature	Testing	Inspection
Purpose	To detect functional defects or performance issues	To detect nonconformance to requirements
Method	Execute the product (run code, simulate behavior)	Examine artifacts (documents, code, plans) without execution
Focus	Does it work as expected under certain conditions?	Is it built correctly according to specs?
Output	Test cases, results, bug reports	Review comments, checklist of defects
Performed On	Software, systems, hardware (dynamic)	Documents, code, requirements, designs (static)
When	During or after development	Early in the lifecycle (before or during development)
Examples	Unit testing, integration testing, performance testing	Code review, walkthroughs, document inspections

Outputs

Quality Management Plan

- It describes how the project management team will implement the performing organization's quality policy.
- The quality management plan is a component or a subsidiary plan of the project management plan
- The quality management plan provides inputs to the overall project management plan and must address quality control (QC), quality assurance (QA) and quality management for the project.
- The quality management plan should be reviewed early in the project

QUALITY MANAGEMENT PLAN	
Project Title: _____ Date Prepared: _____	
Quality Standards	
Quality Objectives	
Metric or Specification	Measure
1.	1.
2.	2.
3.	3.
4.	4.
Quality Roles and Responsibilities	
Roles	Responsibilities
1.	1.
2.	2.
3.	3.

Outputs

Quality Metric

- Defines how Quality will be measured
- It can include any type of applicable measurement, including defect rates, bug rates, failure rates, etc.

DELIVERABLE NAME	QUALITY METRIC	MEASUREMENT METHOD
Website average page speed load time	Largest Contentful Paint: less than 2.5s Cumulative Layout Shift: less than 0.1s	Google Page Speed Insights
User Acceptance Testing	Defects found during UAT less than: 5 - critical 10 - High 20 - Medium	Defects found during UAT will be logged on the Defect Tracking Log. The project manager will use it to calculate the number of defects that fall into each category.
Telephony routing	Deflect 30% of calls within 3 months of launch.	Telephony tool reporting
User Acceptance Testing	UAT to complete before code freeze on 10 December.	Project schedule (tracking gantt)

Manage Quality

Manage Quality

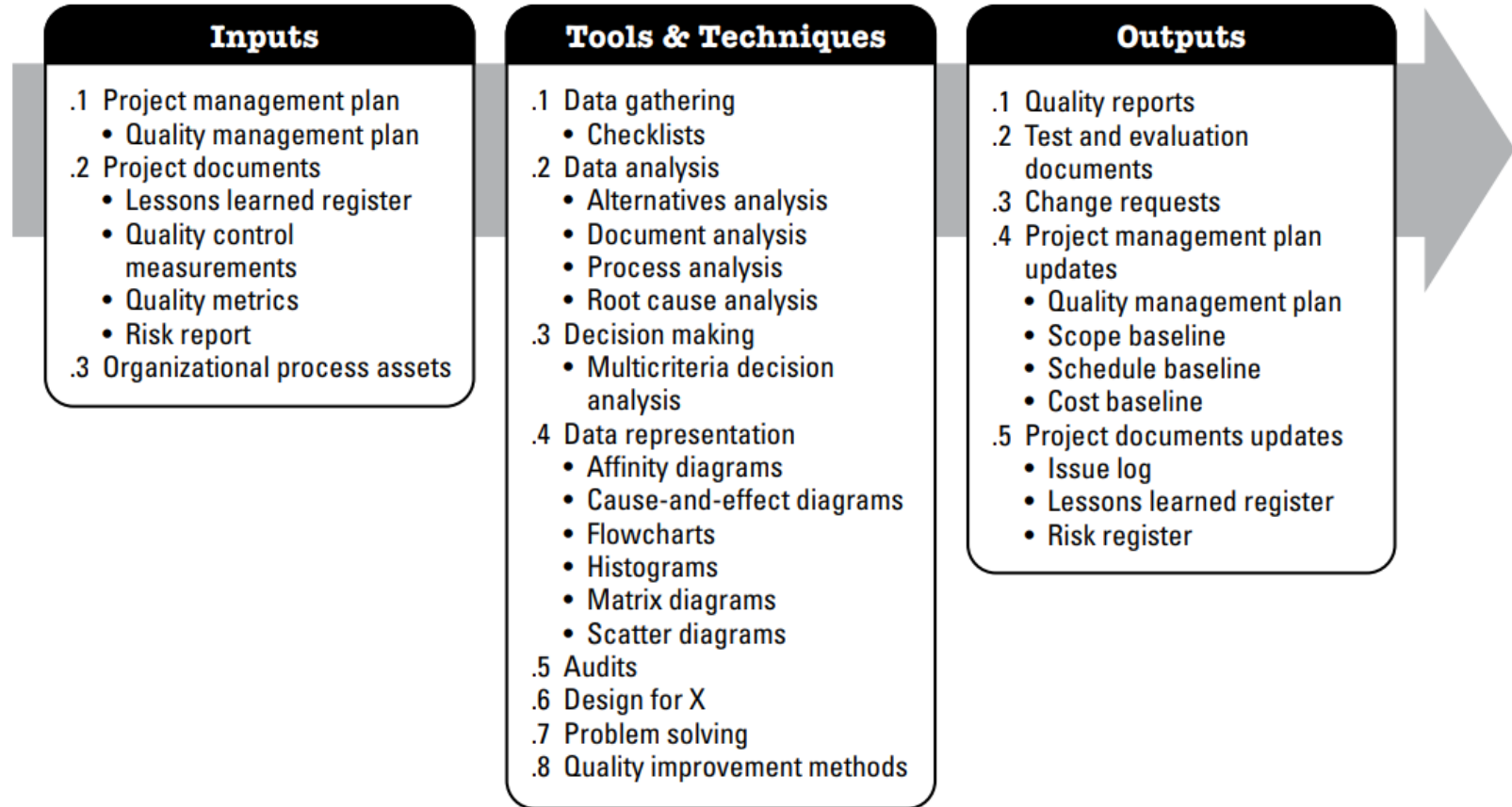
- **What:** process of translating the quality management plan into executable quality activities that incorporate the organization's quality policies into the project.
- **Why:**
 - Increases the probability of meeting the quality objectives
 - Identifying ineffective processes and causes of poor quality
- **When:** This process is performed throughout the project.
 - Quality planning should be performed in parallel with the other planning processes.
- **How:** uses the data and results from the control quality process to reflect the overall quality status of the project to the stakeholders

Manage Quality

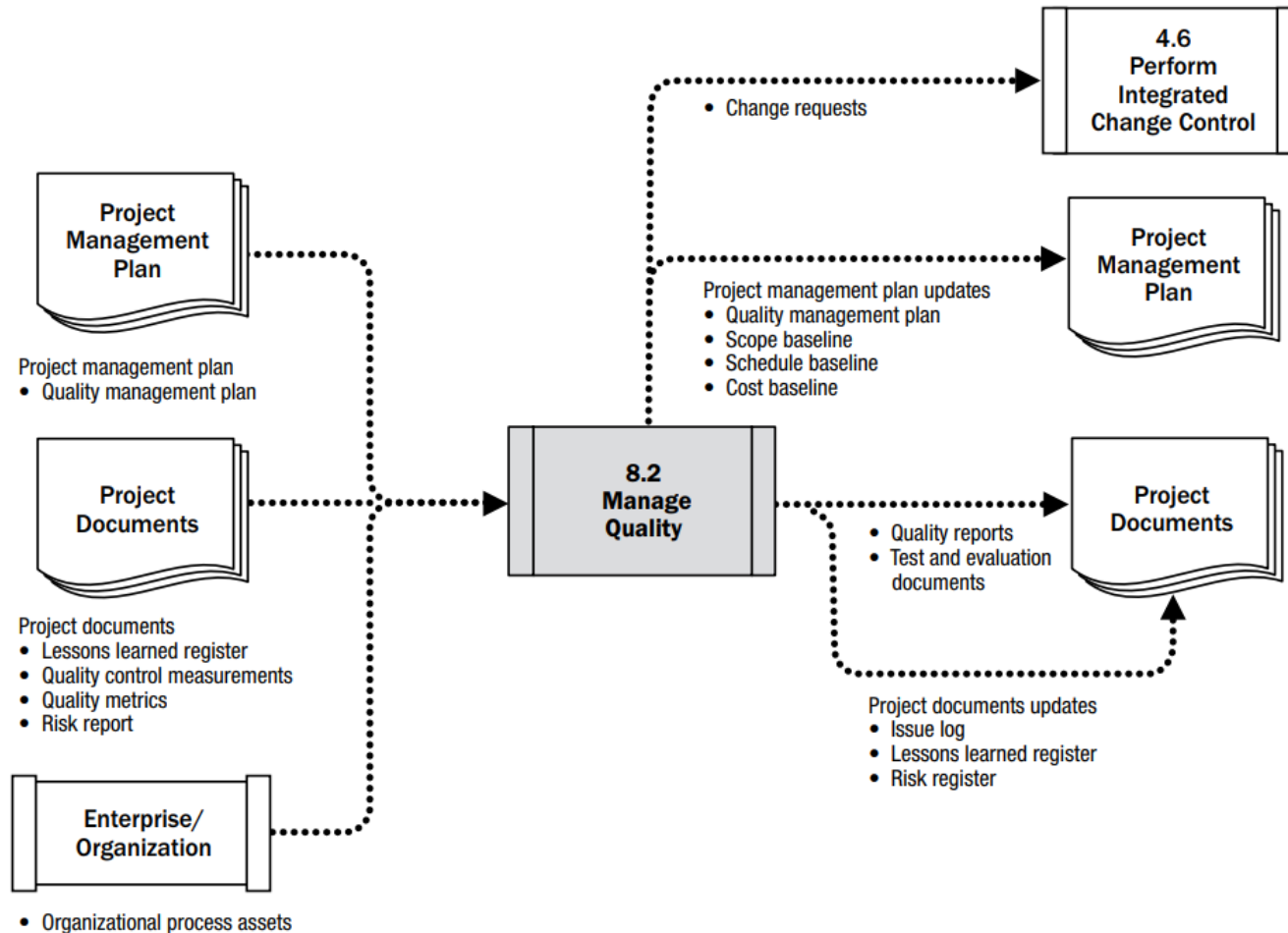
- **How:** Manage Quality is sometimes called quality assurance, although Manage Quality has a broader definition than quality assurance as it is used in nonproject work (including product design aspects and process improvements).
 - It uses the data and results from the control quality process to reflect the overall quality status of the project to the stakeholders.
- Manage Quality is considered the **work of everybody**—the project manager, the project team, the project sponsor, the management of the performing organization, and even the customer.
 - Organization's quality assurance department or other organizational functions may execute some of the Manage Quality activities such as failure analysis, design of experiments, and quality improvement.



Process



Data Flow



Tools & Techniques: Data-gathering

Checklist: a structured tool, usually component-specific, used to verify that a set of required steps has been performed or to check if a list of requirements has been satisfied.

- Quality checklists are often used to ensure consistency in frequently performed tasks.
- It may be simple or complex phrased as imperatives (do this) or Interrogatories (Have you done this) in which process you may ask it
- Quality checklists should incorporate **the acceptance criteria** included in the **scope baseline**.

Quality Checklist					
Project:				Date:	
	Verification				
Quality Item	Yes	No	N/A	Date	Comments
Does the project have an approved quality management plan?					
Has the quality management plan been reviewed by all stakeholders?					
Do all stakeholders have access to the quality management plan?					
Is the quality management plan consistent with the rest of the overall project plan?					
Have product quality metrics been established, reviewed, and agreed upon?					
Have process quality metrics been established, reviewed, and agreed upon?					
Do all metrics support a quality standard which is acceptable to the customer?					
Do all metrics have agreed upon collection mechanisms?					

Tools & Techniques: Data analysis

Alternatives analysis.

This technique is used to evaluate identified options in order to select which different quality options or approaches are most appropriate to use.



Document analysis

The analysis of different documents produced as part of the output of project control processes, such as **quality reports**, **test reports**, **performance reports**, and variance analysis, can point to and focus on processes that may be out of control and may jeopardize meeting the specified requirements or stakeholders' expectations.



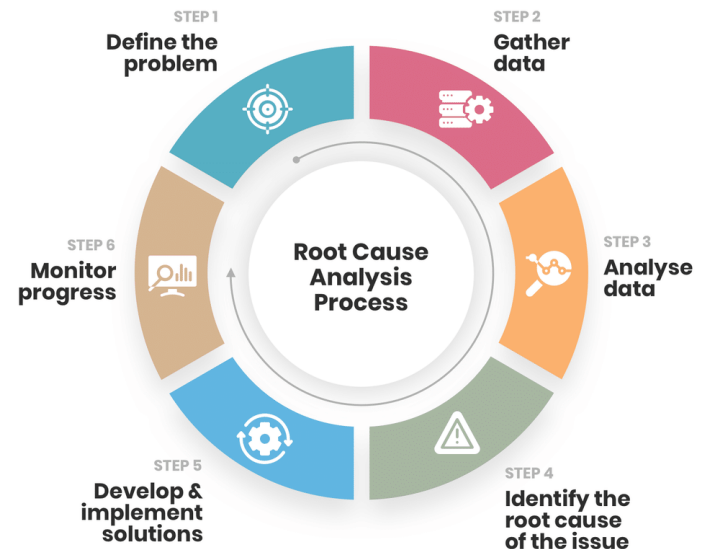
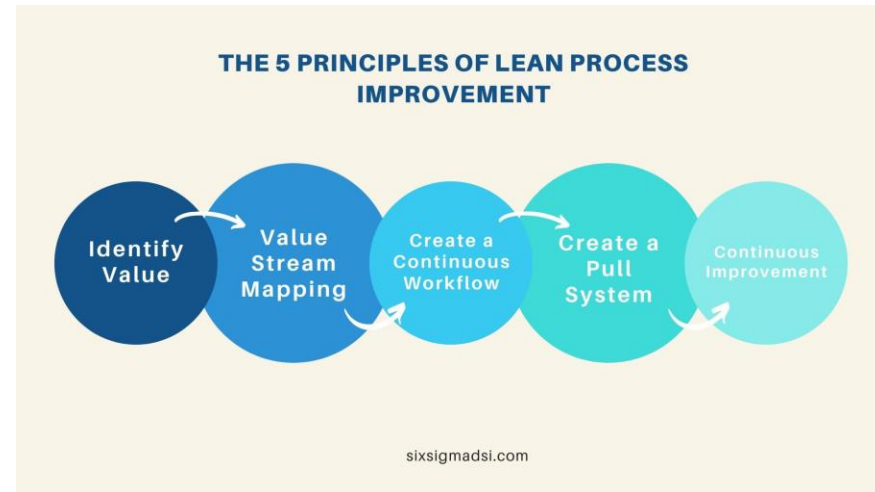
Tools & Techniques: Data analysis

Process analysis

Process analysis identifies opportunities for process improvements. This analysis also examines problems, constraints, and non-value-added activities that occur during a process.

Root cause analysis (RCA)

Root cause analysis is an analytical technique used to determine the basic underlying reason that causes a variance, defect, or risk. A root cause may underlie more than one variance, defect, or risk. It may also be used as a technique for identifying root causes of a problem and solving them. When all root causes for a problem are removed, the problem does not recur.



Tools & Techniques: Data Representation

Affinity diagrams

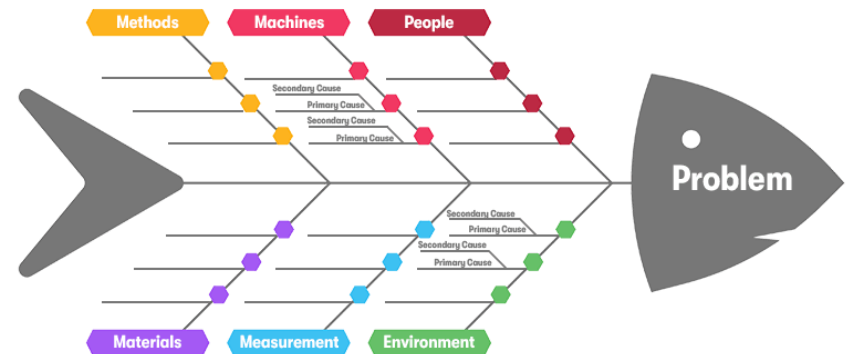
Affinity diagrams can organize potential causes of defects into groups showing areas that should be focused on the most.



Cause-and-effect diagrams

(Also known as fishbone diagrams, why-why diagrams, or Ishikawa diagrams)

This type of diagram breaks down the causes of the problem statement identified into discrete branches, helping to identify the main or root cause of the problem.



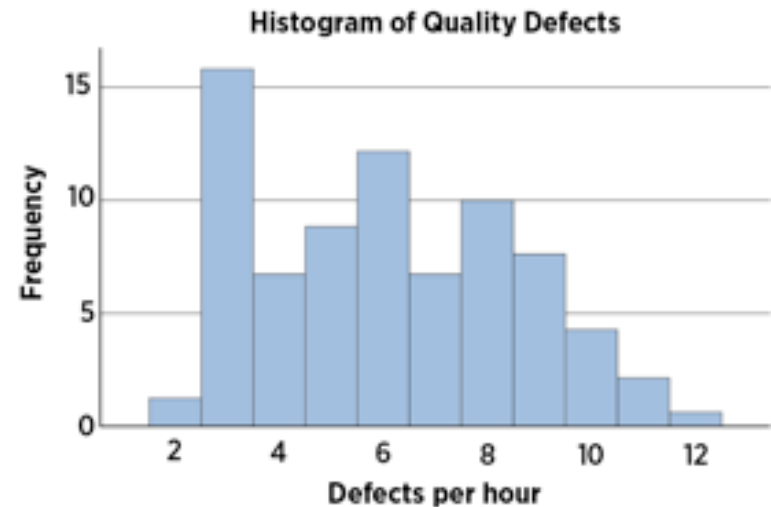
Tools & Techniques: Data Representation

Flowcharts

Flowcharts show a series of steps that lead to a defect.

Histograms

Histograms can show the number of defects per deliverable, a ranking of the cause of defects, the number of times each process is noncompliant, or other representations of project or product defects .



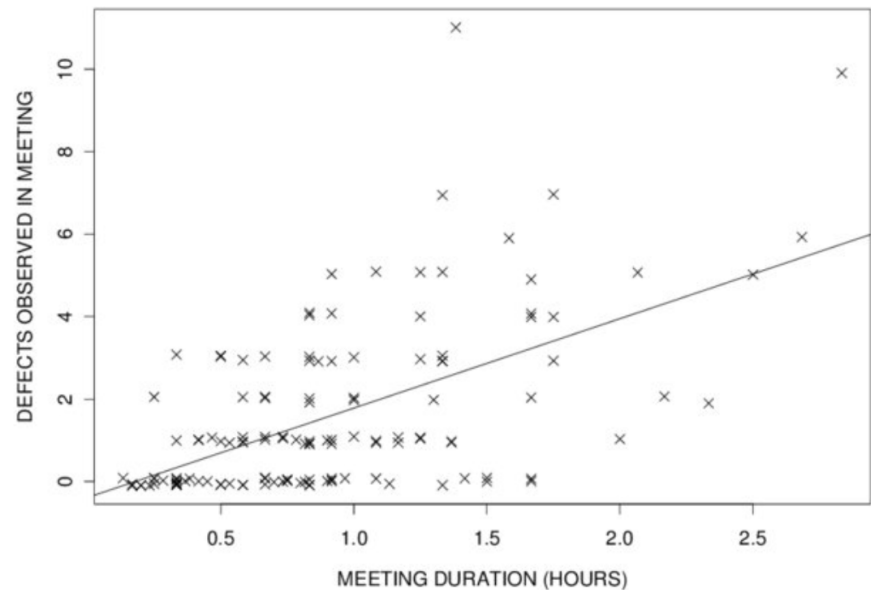
Tools & Techniques: Data Representation

Matrix diagrams

The matrix diagram seeks to show the strength of relationships among factors, causes, and objectives that exist between the rows and columns that form the matrix.

Scatter diagrams

Scatter diagrams can demonstrate a relationship between any element of a process, environment, or activity on one axis and a quality defect on the other axis.



Tools & Techniques: Audit

An audit is a structured, independent process used to ***determine if project activities comply with*** organizational and project policies, processes, and procedures.

A quality audit is usually conducted by a team external to the project, such as the organization's internal audit department, PMO, or by an auditor external to the organization.

Quality audit objectives may include but are not limited to:

- Identifying all good and best practices being implemented;
- Identifying all nonconformity, gaps, and shortcomings;
- Sharing good practices introduced or implemented in similar projects in the organization and/or industry;
- Proactively offering assistance in a positive manner to improve the implementation of processes to help raise team productivity;
- Highlighting contributions of each audit in the lessons learned repository of the organization.



Tools & Techniques: Design for X

Design for X (DfX) is a ***set of principles and techniques*** used in product and process design where "X" represents a specific focus area (like quality, reliability, cost, manufacturability, safety, etc.).

- A systematic approach to optimize a product or process based on a particular objective (the "X").
- Ensures that key factors are considered early in the design phase.
- Widely used in engineering, manufacturing, and project quality management.



Design for Quality

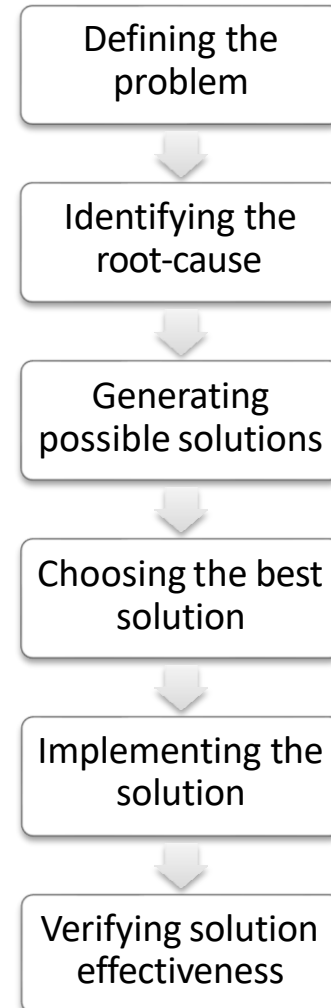
A team developing a mobile app applies *Design for Quality* by:

- Defining clear **code quality standards** (e.g. code coverage, cyclomatic complexity).
- Designing **unit tests and automated test cases** early in the development.
- Selecting frameworks that **enforce modular and testable architecture** (e.g., MVC).
- Using **static code analysis tools** integrated into CI/CD pipeline.
- Applying design patterns that **reduce bugs** (e.g., Singleton, Observer).

Tools & Techniques

Problem solving

- Problems can arise as a result of the Control Quality process or from quality audits and can be associated with a process or deliverable.
- Using a structured problem-solving method will help eliminate the problem and develop a **long-lasting solution**.



Tools & Techniques

Decision making

Multicriteria decision making is used to evaluate several criteria when discussing alternatives that impact project or product quality.

- ***Project decisions*** can include choosing among different implementation scenarios or suppliers.
- ***Product decisions*** can include evaluating the life cycle cost, schedule, stakeholder satisfaction, and risks associated with resolving product defects.



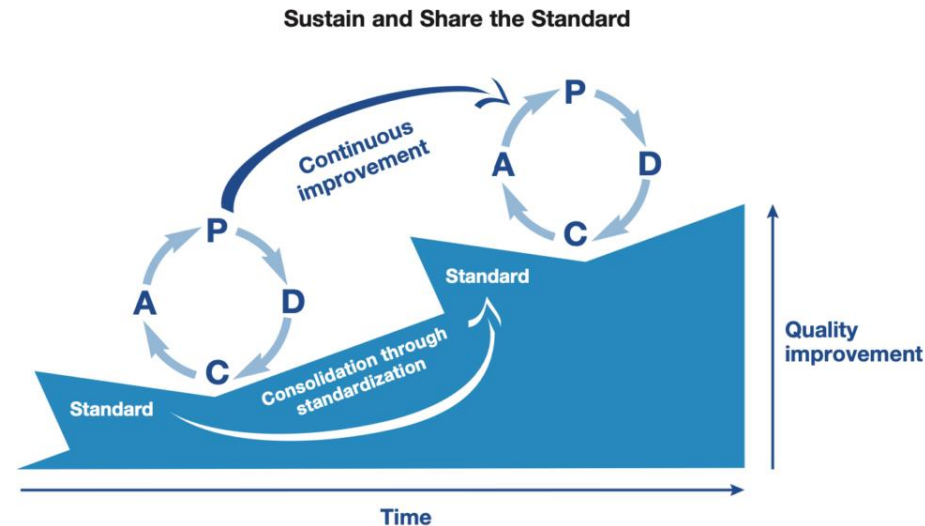
Quality improvements methods

Plan, Do, Check, Act (PDCA)

PDCA is an improvement cycle based on the scientific method of proposing a change in a process, implementing the change, measuring the results, and taking appropriate action.

PDCA is the foundation of continuous improvement or kaizen.

- **Plan** — determine goals for a process and needed changes to achieve them.
- **Do** — implement the changes.
- **Check** — evaluate the results in terms of performance
- **Act** — standardize and stabilize the change or begin the cycle again, depending on the results



改善
kaizen

Quality improvements methods

Six Sigma

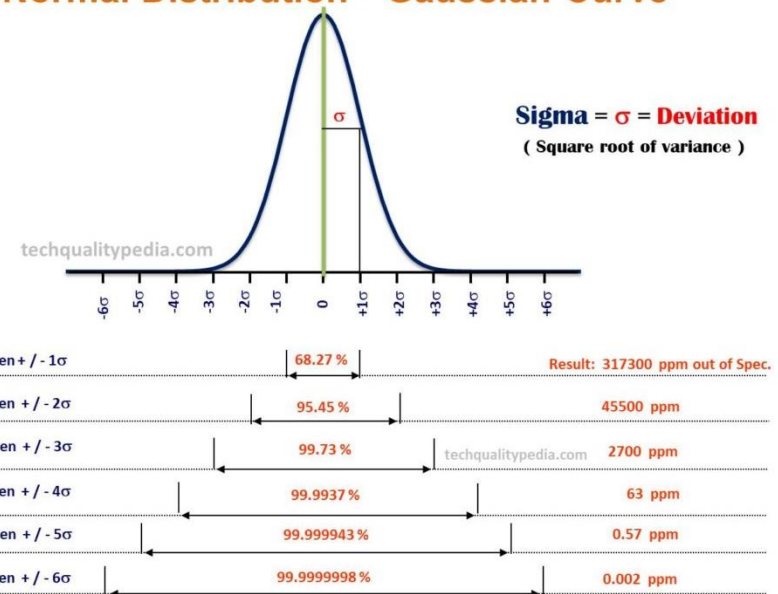
Six Sigma is an approach to quality management that focuses on **minimising the defects** in a product or service by measuring the number of defects in a process and systematically resolving them.

- Its emphasis lies on prevention, continuous monitoring, and data collection.
- The main value of the approach is that it focuses on evidence-based decision-making processes, which are backed up by data and information.
- The two main methods for Six Sigma are the DMAIC method and DMADV method

A Six Sigma process has a **very low probability of defects**.

- 3.4 defects per million opportunities (DPMO)
- 99.99966% defect-free performance

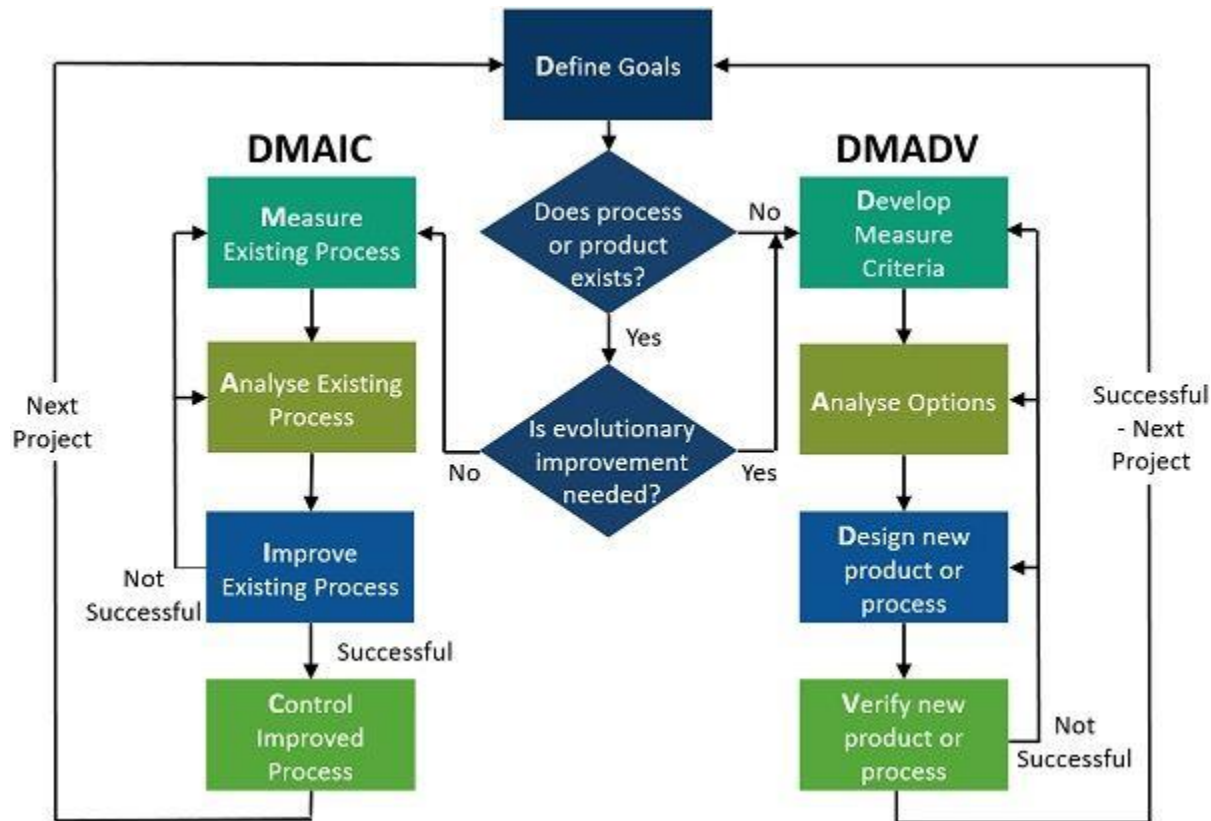
Normal Distribution - Gaussian Curve



Six Sigma

BASIS FOR COMPARISON	DEFINE, MEASURE, ANALYSE, IMPROVE AND CONTROL	DEFINE, MEASURE, ANALYSE, DESIGN AND VERIFY
Meaning	DMAIC is a six sigma methodology which attempts to improve the existing process.	DMADV is another six sigma methodology that seeks to design new or existing processes.
Type of Process	Reactive Process	Proactive Process
Focuses on	Process focused	Customer-focused
Team Requirement	Small Team	Large Team
Time Horizon	Short to Medium-term	Long term
Used when	Continuous improvement is required	Reengineering is required
Approach	Corrective	Preventive
Tools used	Statistical and quantitative tools	Qualitative tools
Implemented when	Incremental change is needed	Evolutionary change is needed

Six Sigma



Outputs

Quality reports

The information presented in the quality reports may include:

- All quality management issues escalated by the team;
- Recommendations for process, project, and product improvements;
- Corrective actions recommendations



Test and evaluation documents

These documents may include:

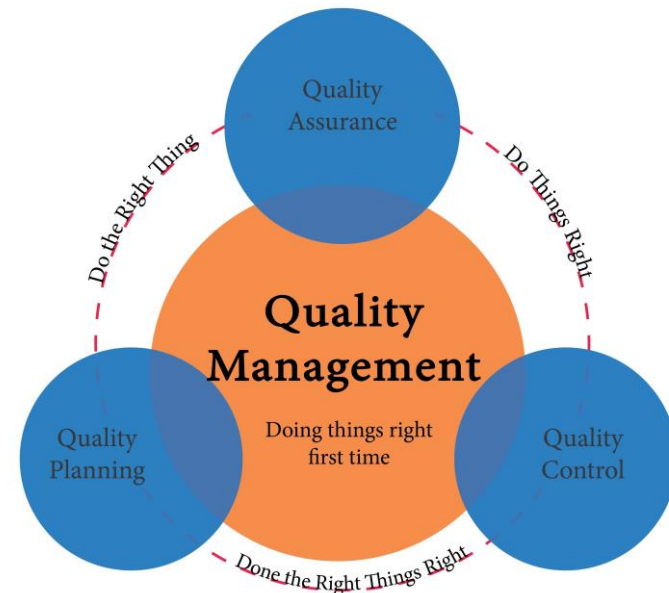
- Dedicated checklists and
- Detailed requirements traceability matrices.
- They are inputs to the Control Quality process and are used to evaluate the achievement of quality objectives.



Control Quality

Control Quality

- **What:** process of monitoring and recording results of executing the quality management activities in order to assess performance and ensure the project outputs are complete, correct, and meet customer expectations.
 - The Control Quality process determines if the project outputs do what they were intended to do.
 - Those outputs need to comply with all applicable standards, requirements, regulations, and specifications.
- **Why:**
 - Verifying that project deliverables and work meet the requirements specified by key stakeholders for final acceptance.
- **When:** This process is performed throughout the project.
 - prior to user acceptance and final delivery.

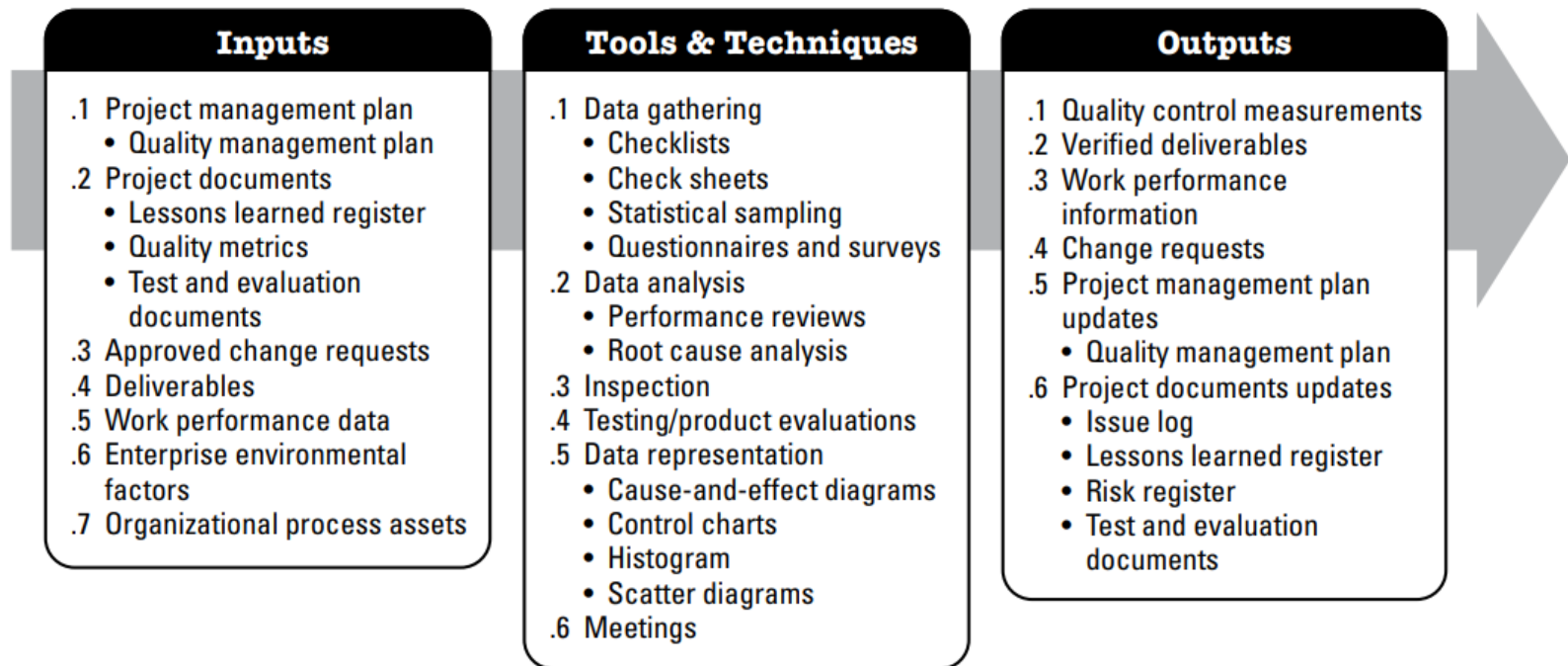


Control Quality

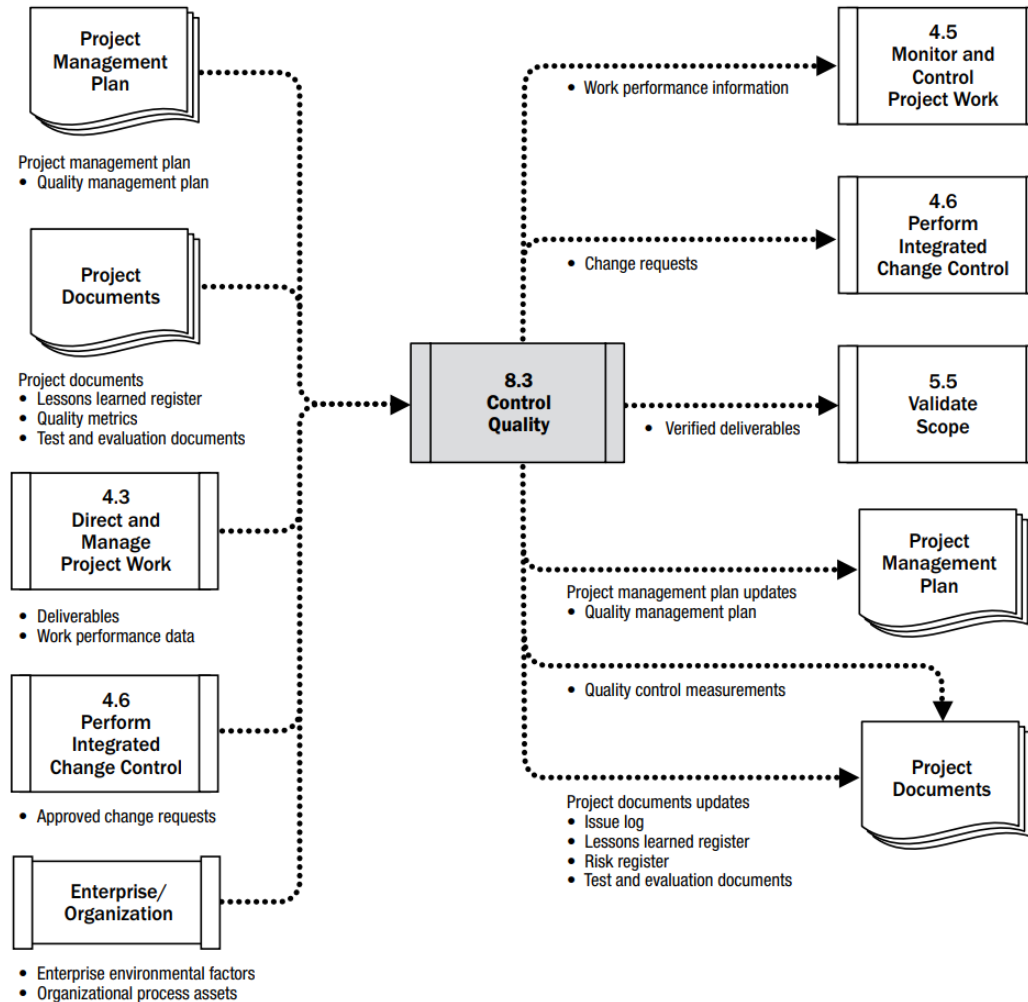
- How:
 - Measure all steps, attributes, and variables used to verify conformance or compliance to the specifications stated during the planning stage
 - Quality control should be performed throughout the project to formally demonstrate, with reliable data, that the sponsor's and/or customer's acceptance criteria have been met.
- The level of effort to control quality and the degree of implementation may differ between industries and project management styles
 - Agile projects: the Control Quality activities may be performed by all team members throughout the project life cycle.
 - Waterfall model-based projects: the quality control activities are performed at specific times, toward the end of the project or phase, by specified team members.



Process



Data Flow

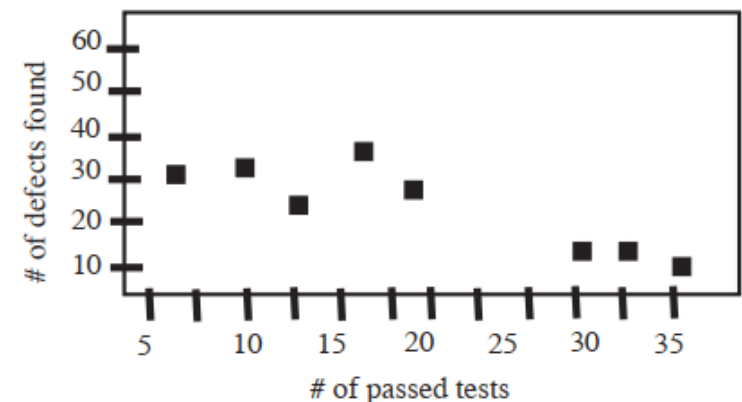


Tools & Techniques: Data-gathering

Check sheets. Check sheets are also known as tally sheets and are used to organize facts in a manner that will facilitate the effective collection of useful data about a potential quality problem. They are especially useful for gathering attributes data while performing inspections to identify defects; for example, data about the frequencies or consequences of defects collected.

Defects/Date	Date 1	Date 2	Date 3	Date 4	Total
Small scratch	1	2	2	2	7
Large scratch	0	1	0	0	1
Bent	3	3	1	2	9
Missing component	5	0	2	1	8
Wrong color	2	0	1	3	6
Labeling error	1	2	1	2	6

Statistical sampling is when you look at a representative sample of something to make decisions. For example, you might look at a selection of widgets produced in a factory to figure out which quality activities would help you prevent defects in them. Statistical sampling helps you make decisions about your product without looking at each and every thing you make.



Tools & Techniques: Data-gathering

Questionnaires and surveys can be used to gather feedback from customers when they've had some time to use the product. Sometimes a defect is higher priority to a customer than the team anticipated, and surveys can uncover the customer's perception of a product in a way that otherwise might be difficult for the team to understand.



Meeting

- **Approved change requests review.** review approved change to verify that they were implemented as approved. This review should also check that partial changes are completed and all parts have been properly implemented, tested, completed, and certified.
- **Retrospectives/lesson learned.** A meeting held by a project team to discuss:
 - Successful elements in the project/phase,
 - What could be improved,
 - What to incorporate in the ongoing project and what in future projects, and
 - What to add to the organization process assets.

Tools & Techniques: Data analysis

Root cause analysis (RCA)

is all about evaluating the product your project is creating to understand why defects are occurring. It's not enough to just find the problem, you need to trace it back to the thing that caused it.

5 whys method:

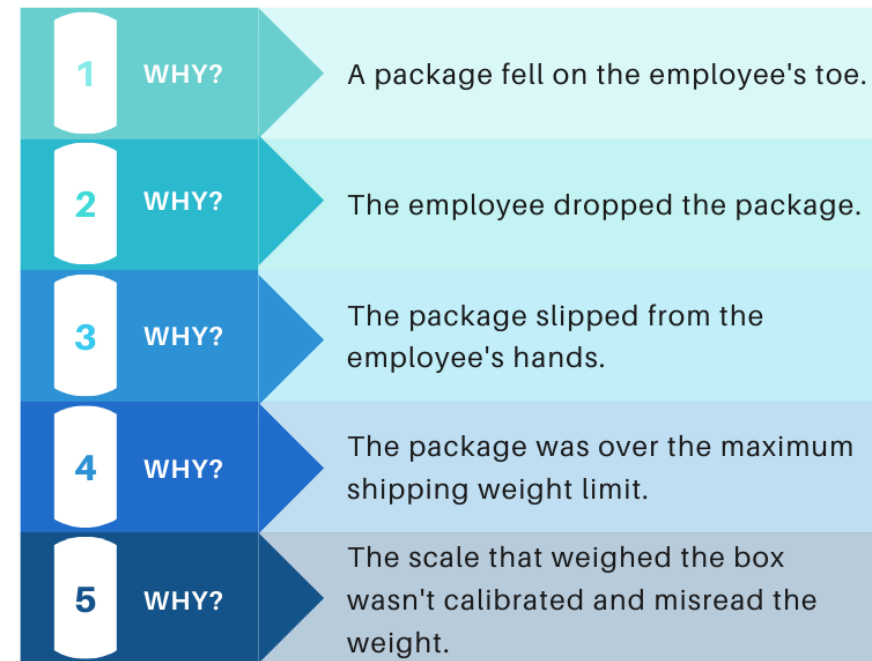
5 why analysis is a technique for finding the root cause of a problem by asking yourself "why did this happen" at least five times.

- Don't stop just because you hit 5. That's arbitrary.
- Don't settle for symptoms or blame (like "user error" or "bad luck") as root causes.
- Avoid "whys" that go in circles or branch off — focus on a single, causal path for clarity.

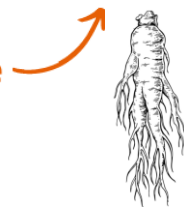
State the Problem:

An employee broke their toe at work.

Ask "Why?" 5 Times:



Identify the Root Cause



Tools & Techniques: Data analysis

Performance reviews are how you and your team will evaluate the current measurements of your product's quality in comparison to the measurements you expected to see.

Performance reviews involve examining actual performance data to:

- Verify compliance with quality requirements.
- Identify variances from standards.
- Support decision-making for improvements.



Manage vs. Control Quality Tools

Tool/Technique	In Manage Quality	In Control Quality
Checklists	Ensure quality standards are embedded in processes.	Verify if steps were followed correctly.
Cause-and-effect diagrams (Ishikawa)	Identify potential causes of future defects (preventive).	Investigate actual defects that occurred.
Statistical sampling	Design quality sampling plans.	Apply sampling to inspect deliverables.
Control charts	Design acceptable variation limits.	Monitor if deliverables are within control.

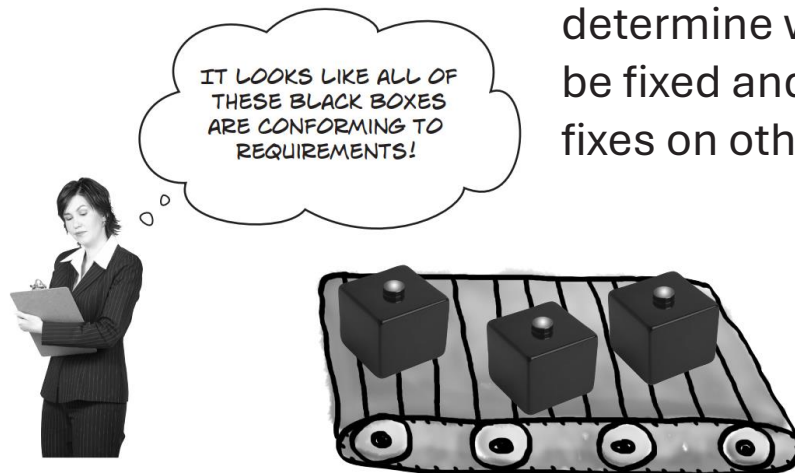
The tools may be similar, but their purpose and timing differ:

- *Manage Quality = "Build it right."*
- *Control Quality = "Check if it's right."*

Tools & Techniques

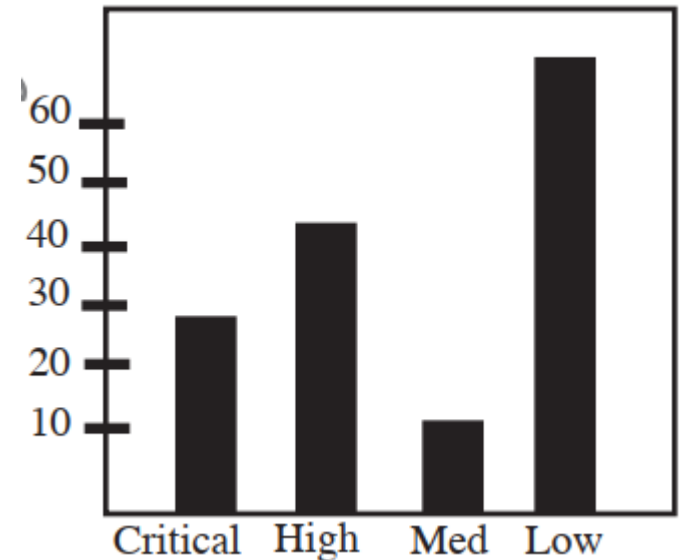
Inspection means comparing the deliverables in your project to the standards you've created for them. If deliverables don't meet your project's standards, you've found a defect. It's not just the product of your project that gets inspected, though. **You can inspect any interim deliverable** as you go—and the sooner you inspect and find defects in a deliverable, the less expensive it is to fix.

Testing/product evaluations are planned as part of the Plan Quality process. As **new deliverables are made** available throughout your project, the team tests them to understand how much they conform to the requirements that have been defined up front. Where the product doesn't behave as specified, that's considered a defect. Usually teams will review all of the defects found during testing to determine whether or not they need to be fixed and the impact of making those fixes on other planned work.

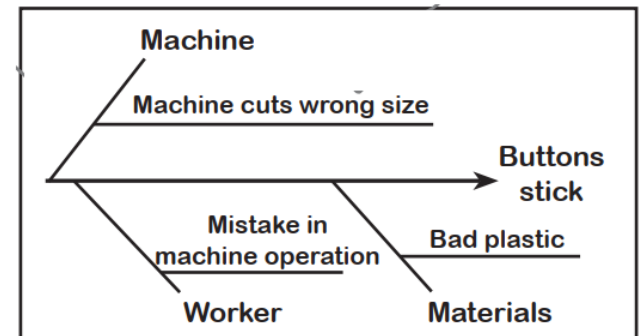


Tools & Techniques: Data Representation

Histograms give you a good idea of how your data breaks down. If you heard that your product had 158 defects, you might think that they were all critical. So looking at a chart like this one would help you to get some perspective on the data. A lot of the bugs are low priority. It looks like only 28 or so are critical. Histograms are great for helping you to compare characteristics of data and make more informed decisions



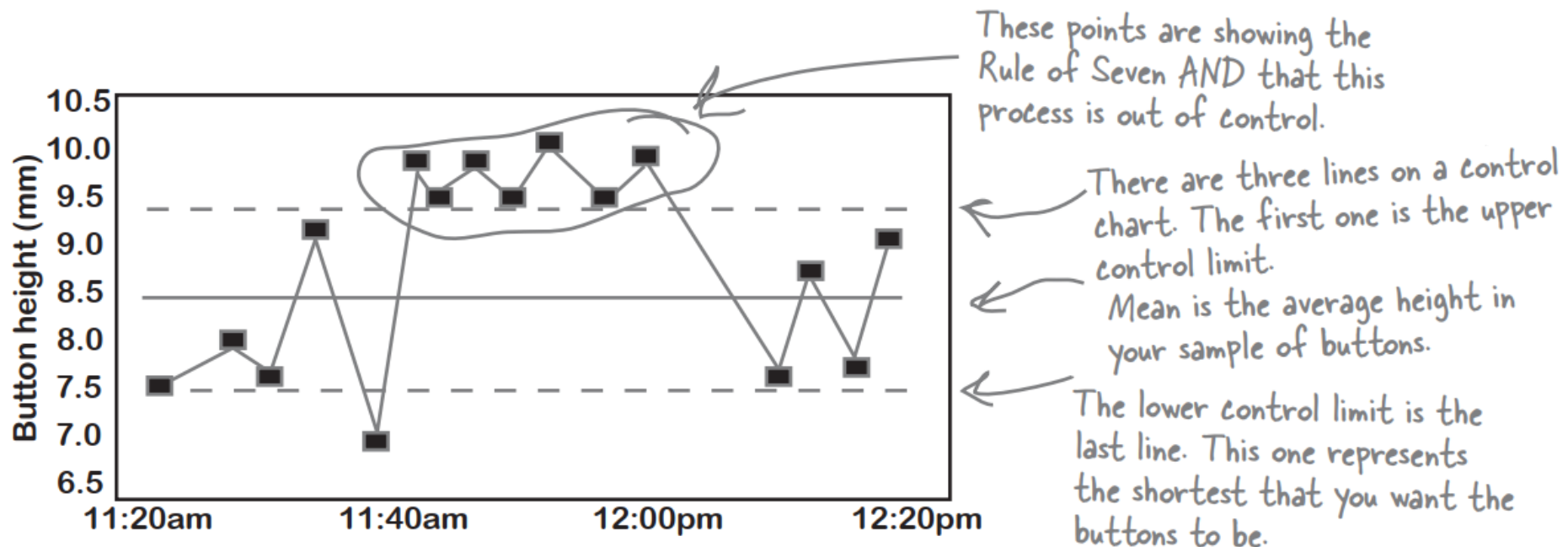
Cause and effect diagrams are also called **fishbone** and **Ishikawa** diagrams. They are used to figure out what caused a defect. You list all of the categories of the defects that you have identified and then write the possible causes of the defect you are analyzing from each category.



Tools & Techniques: Data Representation

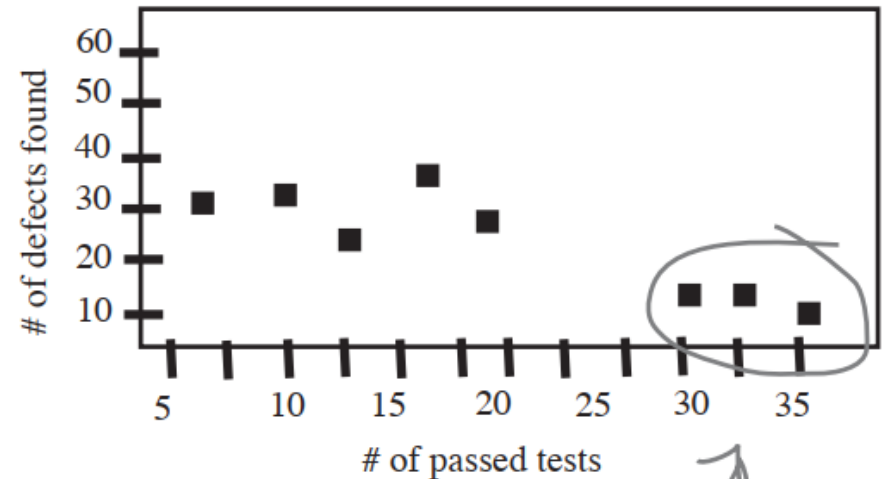
Control charts are a way of visualizing how processes are doing over time.

- When a data point falls outside of the control limits, we say that data point is **out of control**, and when this happens we say that the **entire process is out of control**. It's pretty normal to have your data fluctuate from sample to sample.
- When seven data points in a row fall **on one side of the mean**, that's an uncommon enough occurrence that it means your process might have a problem (**rule of seven**). So when you see this, you need to look into it and try to figure out what's going on.



Tools & Techniques: Data Representation

Scatter diagrams show how two different types of data relate to each other. If you worked with your test team to create a bunch of new tests, you might use a scatter diagram to see if the new test cases had any impact on the number of defects you found. The chart here shows that as more test cases pass, fewer defects are found.



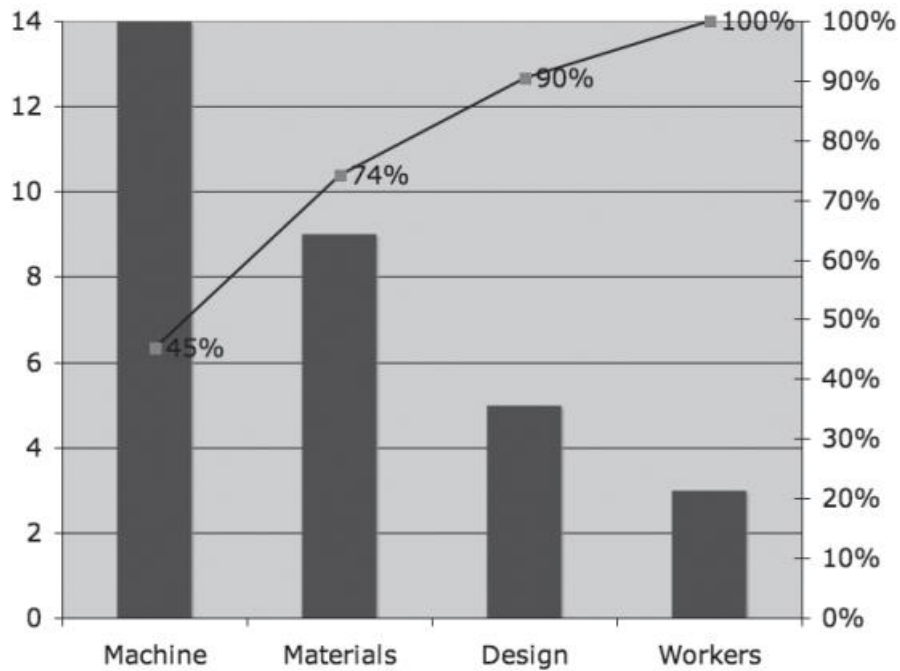
When the number of tests passing goes up, fewer defects are found.

Inspection vs. Testing

Aspect	Inspection	Testing
Definition	A static, manual quality control process where software artifacts (code, design, documents) are reviewed to find defects.	A dynamic process of executing the software to find runtime bugs and validate functionality.
Focus	Process compliance and defect prevention	Functional correctness and defect detection
When	Early in the lifecycle (e.g., during design or coding)	After implementation, during or after integration



"Okay, the motorized rocking chair could use an emergency off switch."



Which root cause is responsible for the most defects in the project?

.....

What is the lowest-priority area for defect prevention?

.....

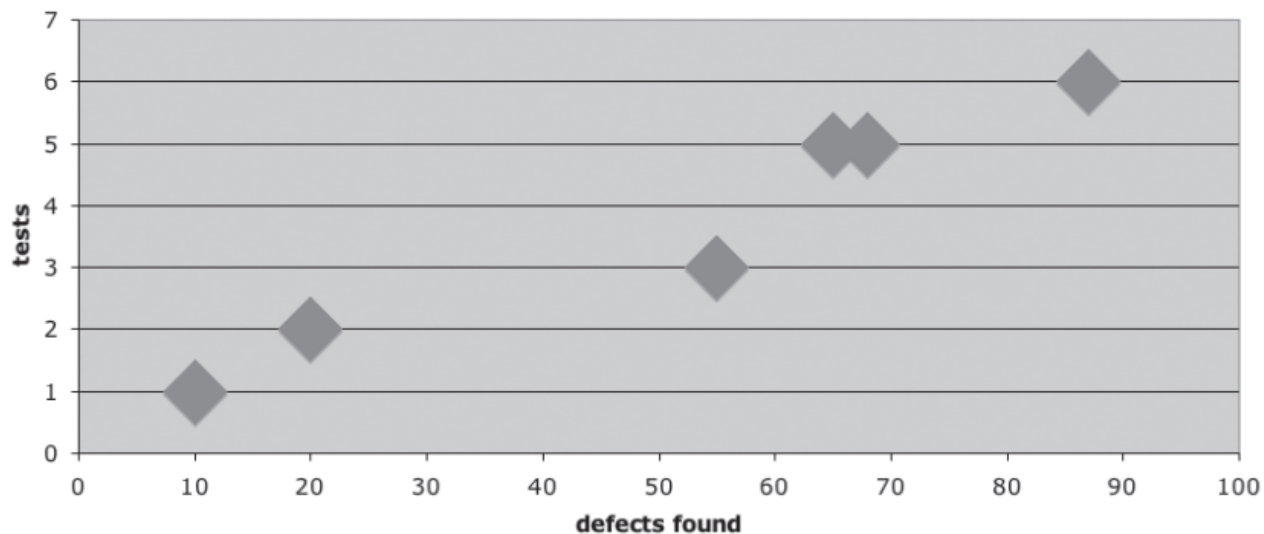
What is the cumulative percentage of machine, materials, and design defects?

.....



Exercise

defects and testers



Hint: Look for a gap in the chart that shows you how adding an extra test caught a lot more defects.

Did adding more tests find more bugs?

.....

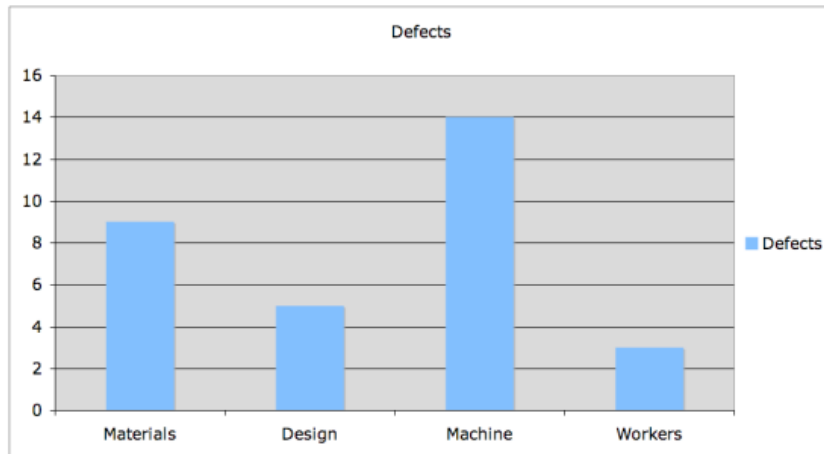
Looking at this chart, should you continue to add more tests to the project?

.....

Where did you see the biggest increase in defect detection?

.....

In other words, did adding extra tests help you find more defects?



How many machine defects were found?

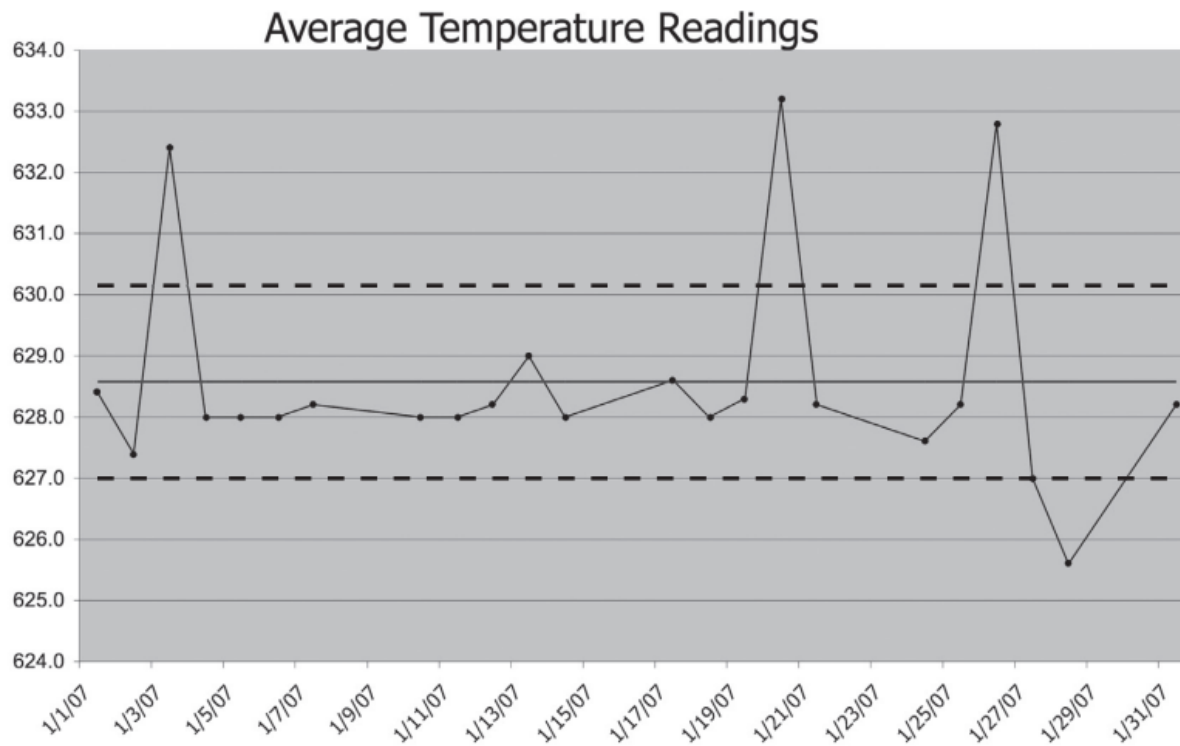
.....

How many defects were caused by workers?

.....

How many total defects are shown on this chart?

.....



Circle the data points that make up the rule of seven.

Is this process in control?

.....

What's the mean temperature reading?

.....

What's the upper control limit?

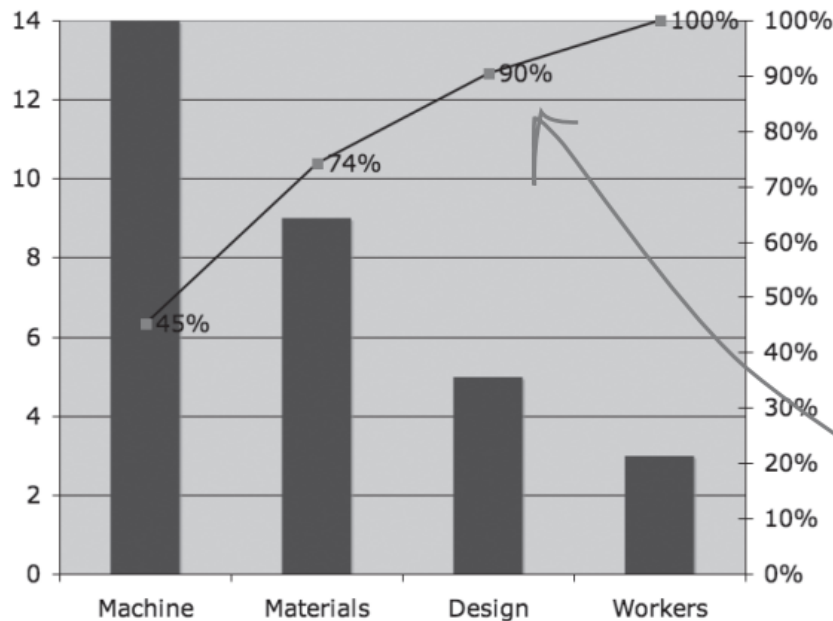
.....

What's the lower control limit?

.....

This is a type of histogram called a Pareto chart that's used to plot out the frequency of defects and sort them in descending order.

You have the most machine defects. So, that's the root cause you should tackle first.



Which root cause is responsible for the most defects in the project?

Machine

What is the lowest-priority area for defect prevention?

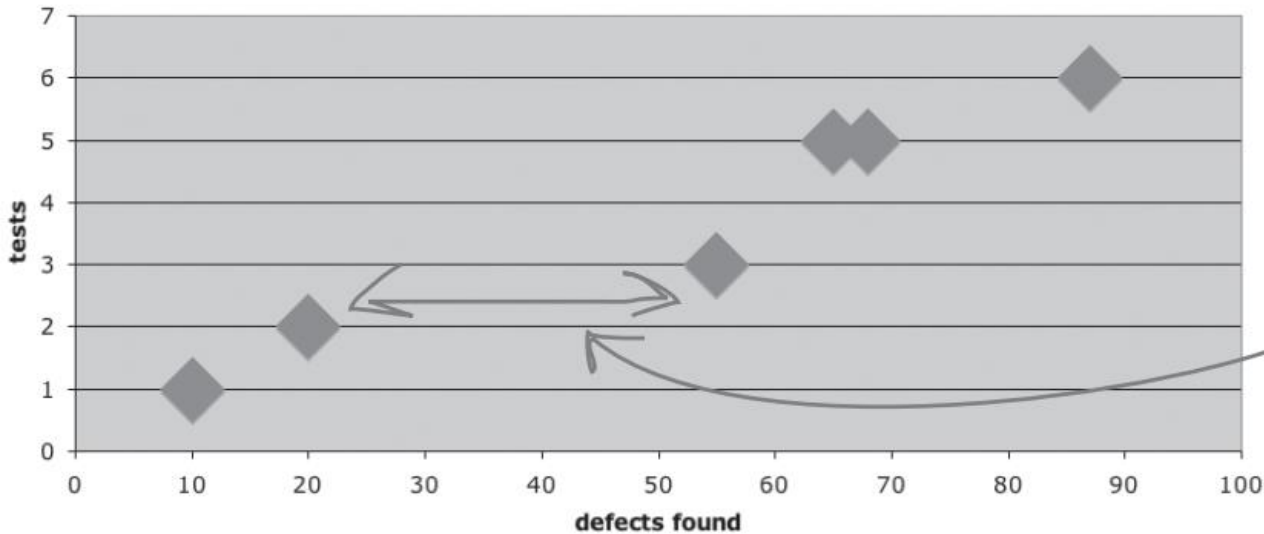
Workers

What is the cumulative percentage of machine, materials, and design defects?

90%

Since you don't have very many worker-related defects, they're the lowest priority for improvement tasks.

defects and testers



The gap between 2 and 3 was the largest, so that's where we had the biggest jump in defect detection.

Did adding more tests find more bugs?

Yes

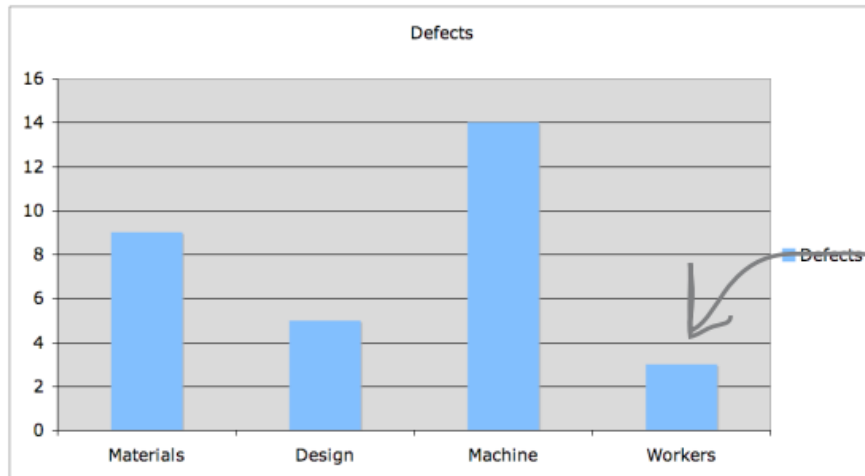
Looks like the number of defects found kept increasing as you added more tests.

Looking at this chart, should you continue to add more tests to the project?

Yes

Where did you see the biggest increase in defect detection?

Three tests



How many machine defects were found?

14

How many defects were caused by workers?

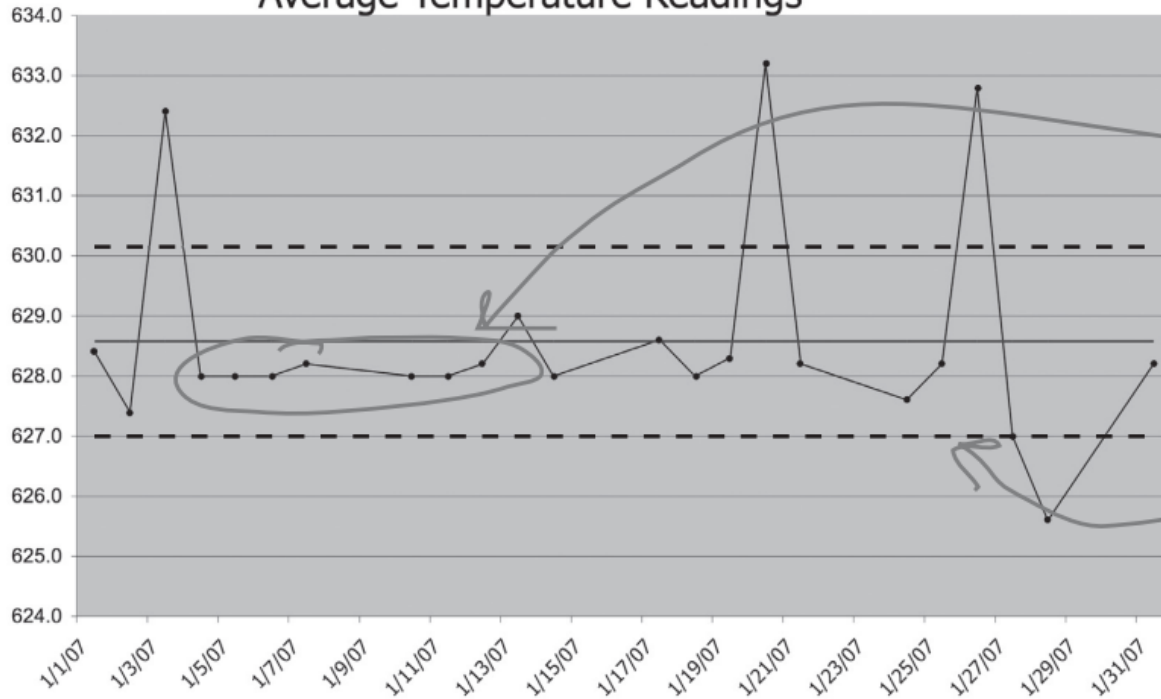
3

How many total defects are shown on this chart?

31

Adding up all of the numbers in the chart tells you how big the dataset is.

Average Temperature Readings



Looks like there's something worth investigating here. There are seven data points in a row that are on the lower side of the mean.

The lower limit is the bottom line and the mean is in the middle.

Circle the data points that make up the rule of seven.

Is this process in control?

No

The points both above and below the control limits tell us that the process is out of control.

What's the upper control limit?

630.2

What's the mean temperature reading?

628.6

What's the lower control limit?

627

Outputs

Verified deliverables and validated changes are two of the most important outputs of Control Quality. Every single deliverable on the project needs to be inspected to make sure it meets your quality standards. If you find defects, the team needs to fix them—and then those repairs need to be checked, to make sure the defects are now gone.

First the team inspects every deliverable to find defects that need to be fixed.



Deliverables

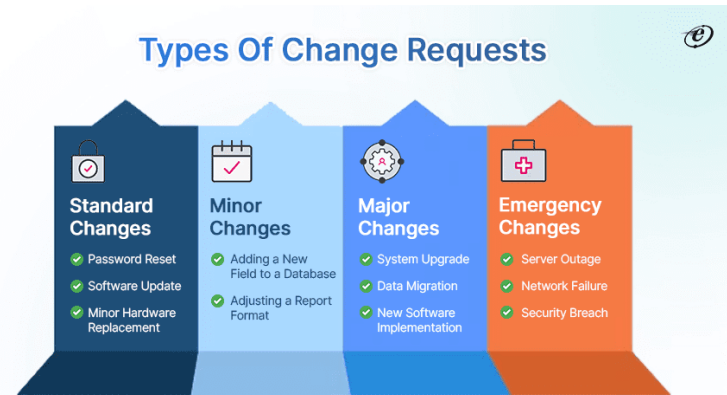


Validated defect repair

When you've finished inspecting your product, you know whether or not your fixes worked.

Outputs

Change requests are recommended or preventive actions that also require changes to the way you are doing your project. Those kinds of changes will need to be **put through change control**, and the appropriate baselines and plans will need to be updated if they are approved.

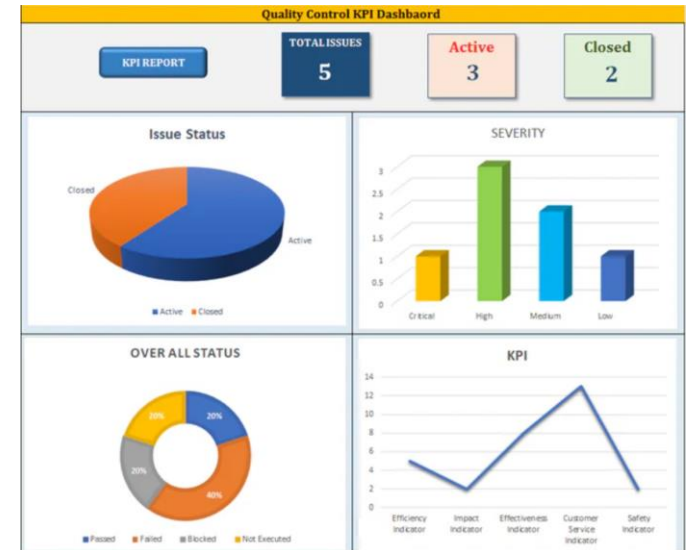


Work performance information might include all of the data your quality processes are producing. Once you've looked at the results of your quality tools, you might find places where the processes you're using to build your product need to be changed. The data you collect in the Control Quality process can help you make those kinds of changes.



Outputs

Quality control measurements are all of the results of your inspections: the numbers of defects you've found, numbers of tests that passed or failed—stuff like that. You'll use them when you look at the overall process you are using in your company to see if there are trends across projects.



Test and evaluation documents are records of quality activities that are performed through the course of the project and their results. It's a good idea to keep records of the results of reviews and quality tests.

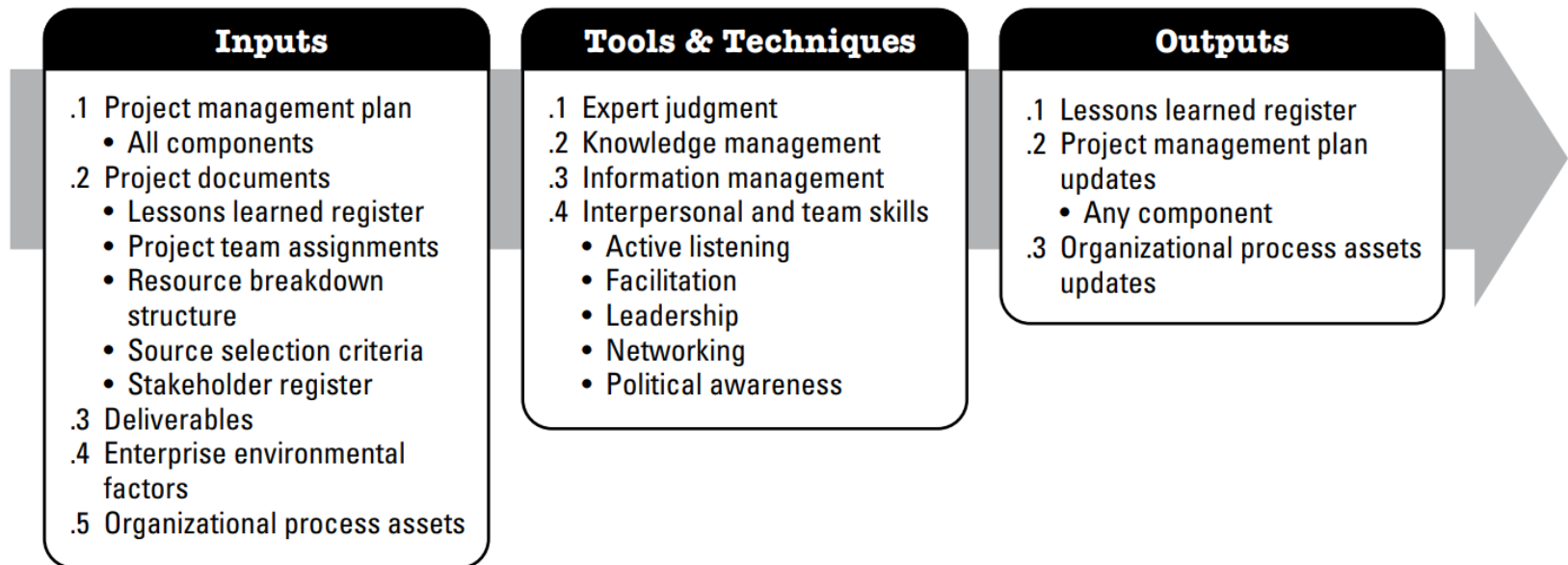


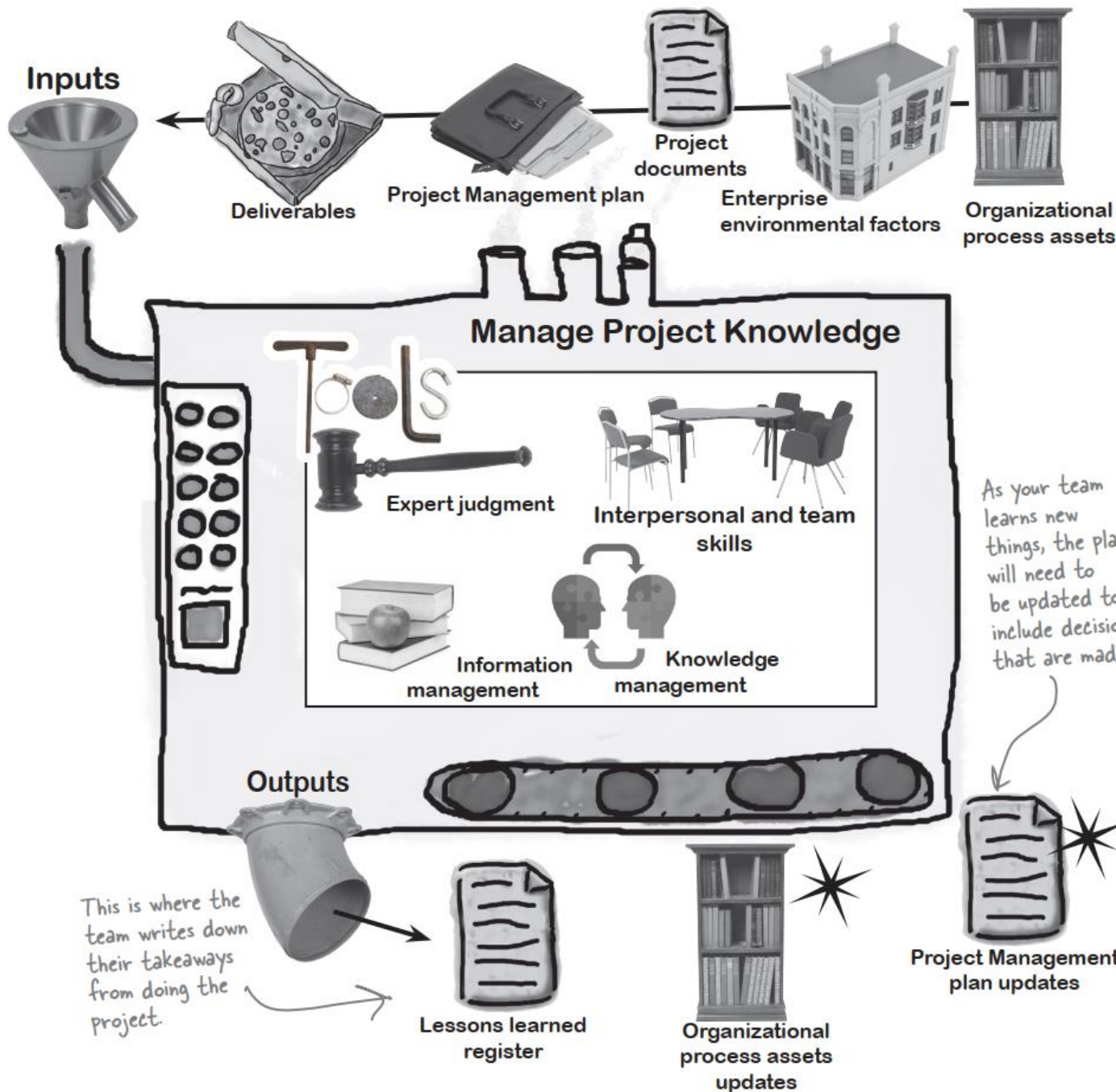
Manage Project Knowledge process

Manage Project Knowledge process

- **What:** the process of
 - using existing knowledge and creating new knowledge to achieve the project's objectives
 - contribute to organizational learning.
- **Why**
 - prior organizational knowledge is leveraged to produce or improve the project outcomes
 - knowledge created by the project is available to support organizational operations and future projects or phases
- **When:** Throughout the project.

Manage Project Knowledge process





using the information your company has already learned from doing past projects and creating new information when your team learns something your company hasn't encountered before

Outputs

Lessons learned register

- The lessons learned register is created as an output of this process early in the project. Thereafter it is used as an **input** and updated as an output in many processes throughout the project.
- Knowledge can be documented using videos, pictures, audios, or other **suitable** means that ensure the efficiency of the lessons captured.
- **The persons or teams** involved in the work are also involved in capturing the lessons learned.
- At the end of a project or phase, the information is transferred to an organizational process asset called a **lessons learned repository**.

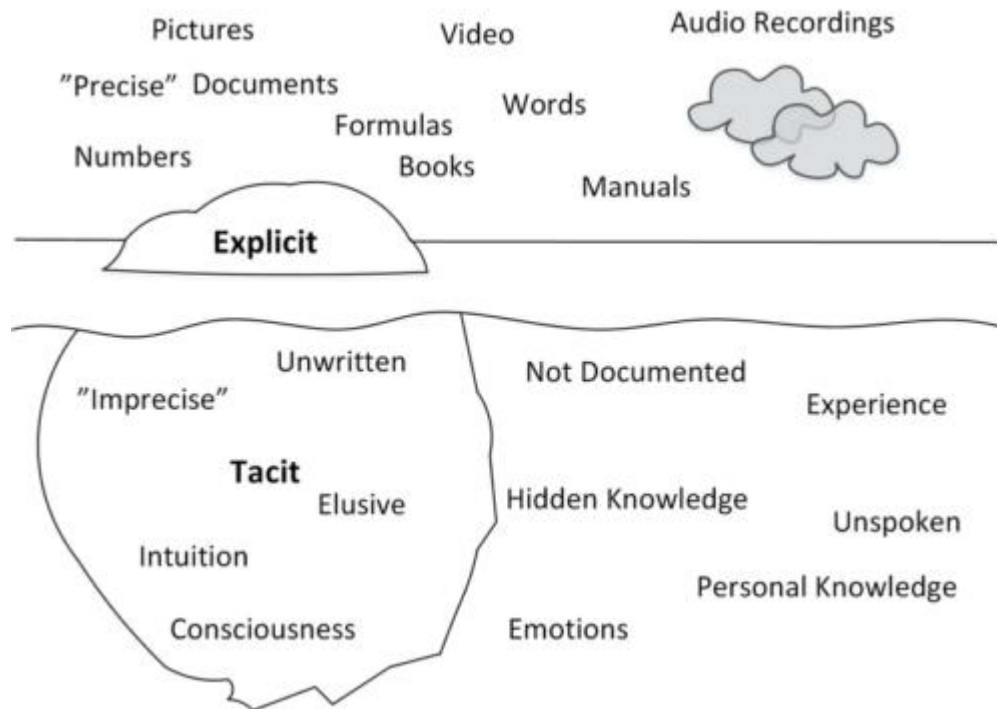
Project management plan updates

- Any component of the project management plan

Organizational process assets updates

- Any organizational process asset can be updated as a result of this process.
- Some of this knowledge is codified, embedded in deliverables, or embedded in improvements to processes and procedures as a result of the Manage Project Knowledge process.

Knowledge is the lifeblood of any project



Tools & Techniques

Knowledge management

Knowledge management tools and techniques connect people so they can work together to create new knowledge, share tacit knowledge, and integrate the knowledge of diverse team members.

- All of these tools and techniques can be applied face-to-face or virtually, or both.
- **Face-to-face interaction** is usually the most effective way to build the trusting relationships that are needed to manage knowledge.
- Once relationships are established, **virtual interaction** can be used to maintain the relationship.

- Networking
- Communities of practice
- Meetings
- Work shadowing
- Online forums
- Seminars
- Conferences;
- Workshops
- Storytelling;
- Knowledge fairs and cafés
- Training

Explicit knowledge:

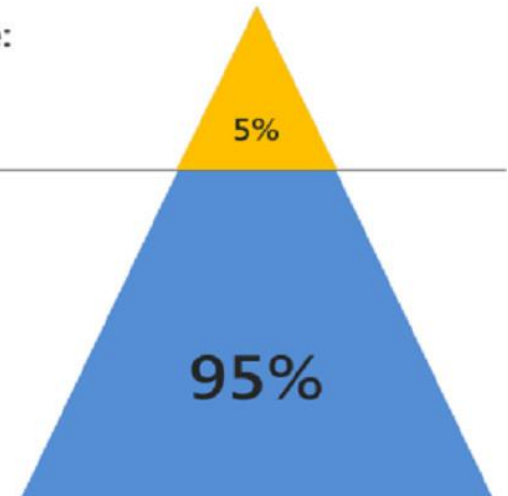
- Data, information
- Documents
- Records
- Files

5%

Tacit knowledge:

- Experience
- Thinking
- Competence
- Commitment
- Deed

95%

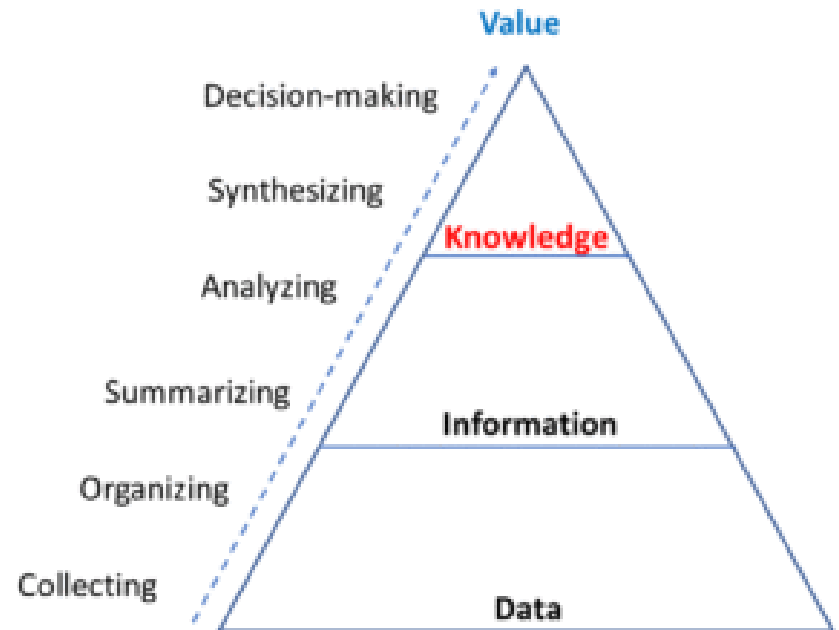


Tools & Techniques

Information management

Information management tools and techniques are used to create and connect people to information. They are effective for sharing simple, unambiguous, **codified explicit knowledge**.

- Methods for codifying explicit knowledge
- Lessons learned register;
- Library services;
- Information gathering, for example, web searches and reading published articles;
- Project management information system (PMIS)



Tools & Techniques

Expert judgment

Expertise should be considered from individuals or groups with specialized knowledge or training in the following topics:

- Knowledge management,
- Information management,
- Organizational learning,
- Knowledge and information management tools, and
- Relevant information from other projects.

Interpersonal and team skills

- **Active listening.** helps reduce misunderstandings and improves knowledge sharing.
- **Facilitation.** helps effectively guide a group to a successful decision, solution, or conclusion.
- **Networking.** allows informal connections and relations among project stakeholders.
- **Leadership.**
- **Political awareness.**

Inputs

Deliverables

- A deliverable is any unique and verifiable product, result, or capability to perform a service that is required to be produced to complete a process, phase, or project.
- Deliverables are typically tangible components completed to meet the project objectives and can include components of the project management plan.

Project management plan

- All components of the project management plan.

Project documents

- **Stakeholder register.** The identified stakeholders to help understand the knowledge they may have.
- **Project team assignments.** The type of competencies and experience available in the project and the knowledge that may be missing.
- **Resource breakdown structure.** Information on the composition of the team and may help to understand what knowledge is available as a group and what knowledge is missing.
- **Lessons learned register.** Information on effective practices in knowledge management.

Inputs

Enterprise environmental factors

- **Organizational knowledge experts.** a team or individual that specializes in knowledge management.
- **Organizational, stakeholder, and customer culture.**
- **Geographic distribution of facilities and resources.** helps determine methods for gaining and sharing knowledge.
- **Legal and regulatory requirements and/or constraints.** These include confidentiality of project information.

Organizational process assets

- **Formal knowledge-sharing and information-sharing procedures.** These include learning reviews before, during, and after projects and project phases
- **Organizational standard policies, processes, and procedures in knowledge management, information management and information security**
- **Personnel administration:** employee development and training records, and competency frameworks

Close Project or a Phase

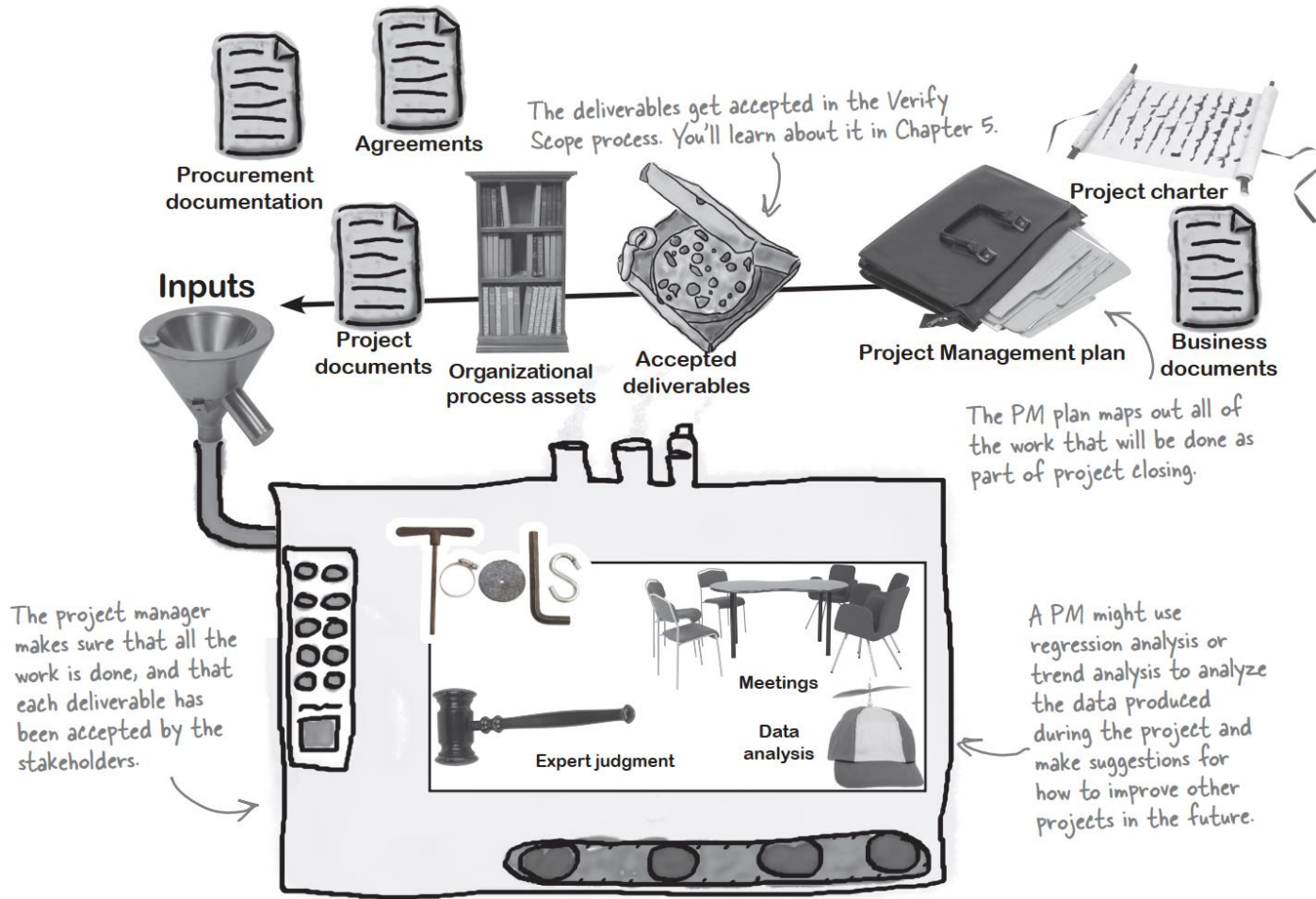
Close Project or a Phase

- **What:** process of finalizing all activities for the project, phase, or contract.
- **Why:**
 - The project or phase information is archived
 - The planned work is completed
 - Organizational team resources are released to pursue new endeavors.
- **When:** performed once or at predefined points in the project

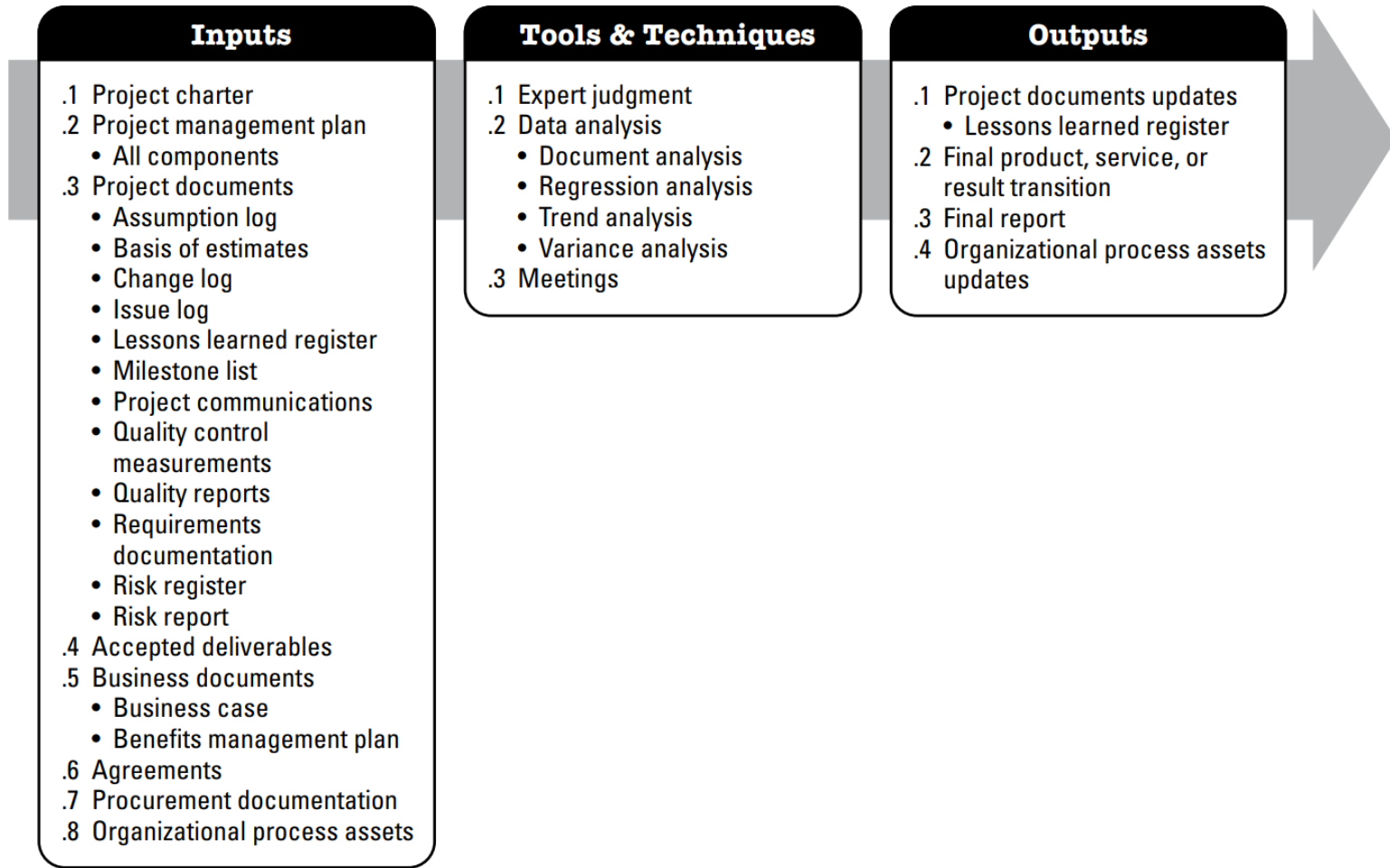
Close Project or a Phase

- **How:** the project manager reviews the project management plan to ensure that **all project work is completed** and that the **project has met its objectives**.
 - Actions and activities necessary to satisfy completion or exit criteria for the phase or project
 - Activities related to the completion of the contractual agreements applicable to the project or project phase
 - Actions and activities necessary to transfer the project's products, services, or results to the next phase or to production and/or operations.
 - Collecting any suggestions for improving or updating the policies and procedures of the organization, and sending them to the appropriate organizational unit.
 - Measuring stakeholder satisfaction.

Process



Process



Tools & Techniques

Expert judgment

Individuals or groups with specialized knowledge or training in the following topics:

- Management control
- Audit
- Legal and procurement
- Legislation and regulations.

Meetings

Types of meetings include but are not limited to:

- Close-out reporting meetings,
- Customer wrap-up meetings,
- Lessons learned meetings, and
- Celebration meetings.

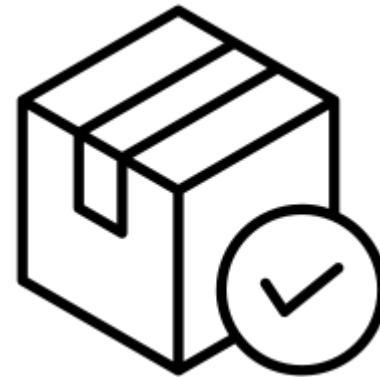
Data analysis

- **Document analysis.** Assessing available documentation will allow identifying lessons learned and knowledge sharing for future projects and organizational assets improvement.
- **Regression analysis.** This technique analyzes the interrelationships between different project variables that contributed to the project outcomes to improve performance on future projects.
- **Trend analysis.** Trend analysis can be used to validate the models used in the organization and to implement adjustments for future projects.
- **Variance analysis.** Variance analysis can be used to improve the metrics of the organization by comparing what was initially planned and the end result.

Outputs

Final Product, Service, or Result Transition

- This output refers to this **transition** of the final product, service, or result that the project was authorized to produce (or in the case of phase closure, the intermediate product, service, or result of that phase) from one team to another, that will operate, maintain, and support it throughout its life cycle.
- This usually requires a **formal sign-off** and, in the case of a project performed on contract, definitely requires a formal sign-off or receipt indicating acceptance of the project.



Outputs

Final report

- The final report provides a summary of the project performance.

Project documents updates

- All project documents may be updated and marked as final versions as a result of project closure.
- Of particular interest is the lessons learned register.

FINAL REPORT

- **Summary description** of the project or phase.
- **Scope objectives:**
 - the criteria used to evaluate the scope, and evidence that the completion criteria were met.
- **Schedule objectives:**
 - the verification and actual milestone delivery dates, and reasons for variances.
- **Cost objectives:**
 - including the acceptable cost range, actual costs, and reasons for any variances.
- **Quality objectives:**
 - the criteria used to evaluate the project and product quality,
- **Summary of the validation** information for the final product, service, or result.